

Trade Tariffs and Exchange Rates: Revisiting Conventional Wisdom in a Three-Country Framework

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Abstract

In the class of models behind the conventional prediction that tariffs appreciate the tariff-imposing country's currency, the exchange rate equals the relative wage and the bilateral terms of trade. A tariff on one trading partner but not another is formally a trade preference in favor of the untariffed partner. The literature on preferential trade agreements, which has studied the third-country effects of such preferences since Viner, therefore applies directly to the exchange rate question. We develop this connection in a three-country model and obtain two results. First, under a single elasticity of substitution across all tradable varieties, the conventional appreciation result survives the move to three countries: for any elasticity and any realistic home bias, a tariff on country B cannot depreciate country A 's currency against the untariffed country C . Second, the result fails once the home-import elasticity η and the cross-origin elasticity ρ are allowed to differ. Country A 's currency depreciates against untariffed partners if and only if $\rho > \rho^* = 3[1 + \alpha_D(\eta - 1) - \alpha_T(1 - \alpha_D)]$. At US-realistic parameters and 2025-scale tariffs the threshold is about 3, within the range of estimated cross-origin elasticities. In a bilateral trade war with similar tariff rates on both sides, both currencies depreciate against the untariffed third country. We calibrate the model to the 2025 US–China episode, comparing each month of data with the tariffs in force in that month. The model matches the April 2025 pattern of modest dollar appreciation against the renminbi and depreciation against the euro. It also shows that the near-uniform tariffs of late 2025 cannot explain the year-end dollar weakness through the trade channel. Observed multilateral depreciation during a tariff episode therefore cannot be attributed to financial channels on the basis of the two-country prediction alone.

Keywords: Trade Tariffs, Exchange Rates, Trade Diversion, Open-Economy Equilibrium

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‡An interactive companion dashboard is available at <https://jasonzhixinglu.github.io/tariff-exchange-rates/>.

1 Introduction

What happens to exchange rates when a country imposes trade tariffs? In practice, exchange rate responses are context dependent and reflect a confluence of factors, but the conventional wisdom derived from a two-country trade model predicts an unambiguous currency appreciation for the tariff-imposing country. The intuition is that tariffs lead to a substitution of expenditure toward domestic goods, and hence an improvement in the bilateral trade balance and an appreciation of the home currency. This result has a long lineage, beginning with [Lerner \(1936\)](#), who established the equivalence between import tariffs and export taxes in a two-country setting and the closely related link between tariffs and exchange rate adjustment under flexible rates. [Metzler \(1949\)](#) extended the analysis to a two-country general equilibrium framework emphasizing the terms-of-trade channel, and [Mundell \(1961\)](#) embedded these ideas in an open-economy macroeconomic framework showing that tariff-induced currency appreciation offsets the trade-balance effects of commercial policy under flexible exchange rates. Modern micro-founded treatments reinforce the same conclusion: [Obstfeld and Rogoff \(1996\)](#) and [Corsetti and Pesenti \(2001\)](#) provided the analytical toolkit for studying policy transmission through terms-of-trade and exchange rate adjustments in fully intertemporal settings with optimizing agents, demonstrating that the relative price channels identified in the classical literature remain operative across a wide range of frameworks.

The way this mechanism works is as follows. A tariff imposed by country A on imports from country B raises the consumer price of B 's tradable goods in country A . Holding the exchange rate fixed, the immediate effect is an expenditure-switching response: households in country A substitute away from imported goods toward domestically produced goods. This contraction in import demand improves country A 's trade balance at the prevailing exchange rate, requiring an appreciation of country A 's currency in equilibrium to restore trade balance. Depending on the treatment of tariff revenue, income effects may reinforce this mechanism by further reducing import demand as real income falls. More generally, any force that lowers import expenditures at a given exchange rate shifts the trade balance condition in the same direction. While additional channels, for example through adjustments in export prices or firm markups, may also be relevant depending on the context, they are not essential for the result.

Empirical evidence from recent tariff episodes has complicated this picture. [Jeanne and Son \(2024\)](#) studies the 2018–19 US-China trade tariffs using a calibrated small open economy model and shows that the model generates only a small appreciation of the USD alongside a large depreciation of the RMB, broadly matching the observed exchange rate pattern during the episode. They also conduct a high-frequency event study showing that US tariff announcements led to an immediate USD appreciation, while Chinese retaliation announcements had no significant exchange rate effects. [Khalil and Strobel \(2024\)](#) argues that tariff-induced increases in trade policy uncertainty contributed to the appreciation of the USD during the same episode, operating through higher safe-haven demand for US Treasuries — an additional channel that reinforces the conventional prediction. The 2025 episode is harder to reconcile with the conventional wisdom. [Cardani et al. \(2025\)](#) studies the April 2025 US reciprocal tariffs using a three-country DSGE open economy

model and documents that the dollar weakened after the announcement, contrary to the conventional prediction. They attribute this to a decline in the convenience yield of US Treasuries, driven by weaker growth expectations and heightened policy uncertainty, which triggered portfolio outflows from the US. Taken together, the 2018–19 episode is broadly consistent with the conventional wisdom, possibly amplified by financial safe-haven flows, while the 2025 episode is not, and the existing literature has turned primarily to financial and convenience-yield channels to account for the gap. Related recent work takes a similar view. Kalemli-Özcan et al. (2025) explain the 2025 dollar depreciation through tariff uncertainty and risk premia, and Werning et al. (2025) study tariffs as cost-push shocks for monetary policy. On the trade side, Darracq Pariès et al. (2026) quantify the third-country spillovers of the 2025 escalation in a three-country New Keynesian model, and Li et al. (2022) estimate the two-elasticity demand structure that we use below. Neither derives the sign result for exchange rates that is our focus. At the same time, the event-study evidence in Corsetti et al. (2025) shows that the exchange rate response depends on the trade configuration: the dollar appreciates when US tariffs are unilateral and depreciates when trading partners retaliate. We show below that real trade channels alone generate this pattern in a multilateral setting.

The starting point of this paper is an observation about what the exchange rate is in the class of models that generates the conventional prediction. With one factor, linear technology, and balanced trade, the nominal exchange rate is proportional to the relative wage measured in a common currency. Because producer prices are unit labor costs, it also equals the bilateral terms of trade. In addition, a tariff imposed by A on B but not on C is formally a trade preference in favor of C : raising τ_{AB} and cutting τ_{AC} create the same discriminatory wedge. It follows that the literature on preferential trade agreements, which has studied the third-country terms-of-trade effects of discriminatory tariffs since Viner (1950) (including Mundell (1964), Kemp and Wan (1976), Bagwell and Staiger (1999, 2002), and the empirical evidence in Chang and Winters (2002)), applies directly to the tariff–exchange-rate question. So does the quantitative literature that computes relative wages under arbitrary tariff vectors (Caliendo and Parro, 2015; Ossa, 2014; Costinot and Rodríguez-Clare, 2014). The contribution of this paper is to make this connection explicit and to work out its implications in a three-country model. Neither literature has stated the implication that matters for the 2025 debate: that a purely real tariff shock can cause the dollar to depreciate against the euro. Section 2.3 develops the equivalence. The analysis is purely real, with no nominal rigidities, no financial frictions, and no uncertainty channels.

When country A imposes a tariff on imports from country B , the expenditure-switching response need not be directed solely toward domestic goods. Expenditure can instead be diverted toward imports from the untariffed third country C , which worsens the bilateral balance between A and C and puts depreciation pressure on A ’s currency. Whether this diversion force can dominate depends on a distinction that a single trade elasticity cannot express. Our first result is an impossibility result: in a three-country CES model in which one elasticity σ governs substitution across all tradable varieties, the conventional appreciation result holds for every value of σ whenever the home share of tradable expenditure is realistic. The reason is that a higher σ strengthens diversion toward C and substitution toward domestic goods in the same proportion, so the diversion force can

never win. We believe this is why the multilateral case has attracted little attention: the natural first extension of the two-country model cannot produce the reversal.

The reversal requires separating the two margins. In a nested demand system with a home-import elasticity η and a cross-origin elasticity ρ , we derive a closed-form threshold, $\rho^* = 3[1 + \alpha_D(\eta - 1) - \alpha_T(1 - \rho)]$, such that an isolated tariff depreciates A 's currency against the untariffed third country if and only if $\rho > \rho^*$. The threshold is increasing in home bias α_D and in η , and decreasing in the tradable expenditure share α_T , in the size of the third country, and in the size of the tariff itself. At US-realistic parameters the threshold is about 4 for small tariffs and about 3 at 2025-scale tariffs, which is within the range of estimated cross-origin elasticities (Feenstra et al., 2018; Broda and Weinstein, 2006; Li et al., 2022; Fajgelbaum et al., 2024). The reversal is therefore empirically plausible, but it cannot appear in any calibration that uses a single elasticity. Finally, in a bilateral trade war the two forces reinforce each other from the third country's perspective, and both currencies depreciate against the untariffed third country, with magnitude increasing in ρ .

The model abstracts from several features present in a more realistic model of multilateral trade, and the results should be interpreted accordingly. First, the three-country framework represents a minimal multilateral environment. Real trade networks involve many countries simultaneously, and the interactions among bilateral exchange rates in a fully multilateral model can be considerably more complex than what we characterize here. Second, the model abstracts from richer production structures, including capital, endogenous labor supply, human capital, intermediate goods, and cross-country technological differences, all of which shape the transmission of tariff shocks in quantitative trade models. Third, the model is static and real: there are no nominal rigidities in consumer prices, no role for invoicing currency, and no intertemporal dynamics. A large literature examines how invoicing currency and nominal price rigidities shape the transmission of tariff shocks to exchange rates and trade flows, which our model does not speak to.¹ Our exchange rate predictions should therefore be interpreted as long-run equilibrium outcomes in which prices have fully adjusted and current account imbalances cannot persist indefinitely.

Introducing short-run dynamics would require nominal frictions, sticky prices, and a financial account that allows trade imbalances to be offset by capital flows. In practice, financial flows are large relative to trade flows and often dominate exchange rate movements at business cycle frequencies, which is one reason observed exchange rate changes may differ substantially from the model's long-run predictions. Given these limitations, the purpose of this paper is not to deliver a realistic framework for quantitative analysis but to establish a conceptual point: one cannot appeal to the conventional two-country wisdom to conclude that any observed depreciation must reflect financial or non-trade factors. In a multilateral setting, real trade channels alone are sufficient to generate depreciation of the tariff-imposing country's currency against untariffed trading partners, and the qualitative direction of the multilateral exchange rate response depends on the structure

¹Devereux and Engel (2003) shows that the currency of price-setting determines exchange rate passthrough, with complete passthrough under producer currency pricing and attenuated passthrough under local currency pricing. Gopinath et al. (2020) extends this framework by introducing a third invoicing currency, and Bergin and Corsetti (2023) show that the resulting asymmetries can create monetary policy tradeoffs that potentially reverse standard predictions about the exchange rate response to tariff shocks.

of trade as much as on financial conditions.

The rest of the paper is organized as follows. Section 2 develops the two-country model, establishes the conventional appreciation result, and shows that its robustness rests on the Marshall–Lerner condition. It also states the equivalence between the exchange rate, the relative wage, and the terms of trade, and explains the scope of the exercise. Section 3 introduces the three-country model, analyzes the three tariff configurations, and presents the two main results: the impossibility result for the single-elasticity model and the reversal threshold ρ^* for the nested model. Section 4 presents a quantitative illustration calibrated to the 2025 US tariff episode. Section 5 concludes.

2 The Two-Country Model

We consider a static two-country, two-sector, one-factor model in which a single nominal exchange rate equilibrates international trade. We first derive the conventional result that tariffs lead to an appreciation of the tariff-imposing country’s currency, obtain a closed-form expression for the equilibrium exchange rate, and then establish the robustness of this result using a general monotonicity argument that does not rely on specific functional form assumptions.

2.1 The Two-Country Conventional Wisdom

Countries are indexed by $i \in \{A, B\}$ and goods by $g \in \{T_A, T_B, N\}$, where T_i denotes the tradable good produced in country i and N denotes a non-tradable good that is produced and consumed domestically. Labor is the only factor of production and is immobile across countries.

2.1.1 Firms

Each good is produced using a linear technology. Output in the tradable and non-tradable sectors satisfies

$$Y_{T_i} = A_{T_i} L_{T_i}, \quad Y_{N_i} = A_{N_i} L_{N_i}, \quad L_{T_i} + L_{N_i} = L_i, \quad (1)$$

where A_{T_i} and A_{N_i} denote sectoral labor productivities and L_i is the total labor endowment in country i . Markets are perfectly competitive. With linear production technologies, firms earn zero profits in equilibrium, and wages in each country are equalized across sectors. We normalize the producer prices of tradable goods to one, so that $P_{T_A} = P_{T_B} = 1$, and maintain this normalization throughout the analysis. Under this normalization, equilibrium wages satisfy

$$w_i = A_{T_i} = P_{N_i} A_{N_i}, \quad (2)$$

which implies $P_{N_i} = A_{T_i}/A_{N_i}$, so that nontradable prices are pinned down by the ratio of sectoral productivities.

2.1.2 Households

Households in each country consume the domestically produced tradable good, the tradable good produced abroad, and a non-tradable good produced locally. Let $C_{T_i i}$ denote country i 's consumption of its own tradable good, $C_{T_j i}$ its consumption of the tradable good produced in the other country $j \neq i$, and C_{N_i} its consumption of the local non-tradable good. Let e denote the nominal exchange rate, defined as units of country A 's currency per unit of country B 's currency, so that the domestic-currency price of T_B in country A is eP_{T_B} and the domestic-currency price of T_A in country B is P_{T_A}/e . Under free trade, household income equals wage income, $I_i = w_i L_i$. The household budget constraints are

$$w_A L_A = P_{T_A} C_{T_{AA}} + eP_{T_B} C_{T_{BA}} + P_{N_A} C_{N_A}, \quad (3)$$

$$w_B L_B = \frac{P_{T_A}}{e} C_{T_{AB}} + P_{T_B} C_{T_{BB}} + P_{N_B} C_{N_B}. \quad (4)$$

Preferences in each country are Cobb–Douglas over the two tradable goods and the local non-tradable good, with expenditure shares α_{T_A} , α_{T_B} , and $\alpha_N = 1 - \alpha_{T_A} - \alpha_{T_B}$ on goods T_A , T_B , and N respectively, where α_{T_j} denotes the common weight on good T_j in both countries' utility functions. Utility in country i is given by

$$U_i = C_{T_A i}^{\alpha_{T_A}} C_{T_B i}^{\alpha_{T_B}} C_{N_i}^{\alpha_N}.$$

Utility maximization subject to the budget constraints implies that households allocate constant expenditure shares across goods. Equilibrium expenditures satisfy

$$P_{T_A} C_{T_A i} = \alpha_{T_A} I_i, \quad P_{T_B}^{(i)} C_{T_B i} = \alpha_{T_B} I_i, \quad P_{N_i} C_{N_i} = \alpha_N I_i, \quad (5)$$

where $P_{T_B}^{(i)}$ denotes the domestic-currency price of T_B faced by country i , equal to eP_{T_B} in country A and P_{T_B} in country B under free trade. These expenditure-share conditions hold generally, with $I_i = w_i L_i$ under free trade; once tariffs are introduced in Section 2.1.4, I_i includes rebated tariff revenue and $I_i \neq w_i L_i$ in general. The corresponding consumption quantities are

$$C_{T_A i} = \frac{\alpha_{T_A} I_i}{P_{T_A}}, \quad C_{T_B i} = \frac{\alpha_{T_B} I_i}{P_{T_B}^{(i)}}, \quad C_{N_i} = \frac{\alpha_N I_i}{P_{N_i}}. \quad (6)$$

2.1.3 Market Clearing and Equilibrium

In equilibrium, markets clear in each country for non-tradable and tradable goods, and international trade is balanced. Non-tradable goods are produced and consumed domestically; for each country $i \in \{A, B\}$, market clearing requires $Y_{N_i} = C_{N_i}$. Using the production technology, the wage condition $P_{N_i} = w_i/A_{N_i}$, and Cobb–Douglas demand, equilibrium labor allocations satisfy

$$\frac{L_{N_i}}{L_i} = \alpha_N, \quad \frac{L_{T_i}}{L_i} = 1 - \alpha_N = \alpha_{T_A} + \alpha_{T_B}. \quad (7)$$

Each tradable variety is produced in only one country and consumed in both. For tradable good T_i , global market clearing requires $Y_{T_i} = C_{T_i i} + C_{T_i j}$, which using (6) gives

$$A_{T_A} L_{T_A} = \frac{\alpha_{T_A} I_A}{P_{T_A}} + \frac{\alpha_{T_A} I_B}{P_{T_A}/e}, \quad A_{T_B} L_{T_B} = \frac{\alpha_{T_B} I_B}{P_{T_B}} + \frac{\alpha_{T_B} I_A}{e P_{T_B}}, \quad (8)$$

where P_{T_A}/e is the price of T_A in country B 's currency and eP_{T_B} is the price of T_B in country A 's currency. Trade balances are measured in each country's own currency; for country A , export revenues are $P_{T_A} C_{T_A B}$ and import expenditures are $eP_{T_B} C_{T_B A}$, giving

$$TB_A = P_{T_A} C_{T_A B} - eP_{T_B} C_{T_B A}. \quad (9)$$

Imposing balanced trade $TB_A = 0$ and substituting (6) yields the equilibrium condition

$$\alpha_{T_A} w_B L_B = \alpha_{T_B} e w_A L_A. \quad (10)$$

By Walras' Law, satisfying (10) for country A ensures global market clearing, so the equilibrium exchange rate is fully determined by a single trade balance condition.

2.1.4 Imposing Tariffs in the Two-Country Model

We introduce an ad valorem tariff $\tau \geq 0$ levied by country A on imports of the tradable good T_B . The tariff raises the consumer price of T_B in country A to $(1 + \tau)eP_{T_B}$, while producers in country B continue to receive the world price $P_{T_B} = 1$, so the tariff wedge τeP_{T_B} is collected as revenue by country A 's government and rebated lump-sum to households. The household budget constraint in country A becomes

$$I_A = P_{T_A} C_{T_A A} + (1 + \tau)eP_{T_B} C_{T_B A} + P_{N_A} C_{N_A}, \quad (11)$$

where $I_A \equiv w_A L_A + T_A^{\text{rev}}$ is total household income and tariff revenue is $T_A^{\text{rev}} = \tau eP_{T_B} C_{T_B A}$. Substituting T_A^{rev} into the definition of I_A and using the Cobb–Douglas expenditure share condition $C_{T_B A} = \alpha_{T_B} I_A / [(1 + \tau)eP_{T_B}]$, income solves to the closed form

$$I_A = \frac{w_A L_A}{1 - \frac{\alpha_{T_B} \tau}{1 + \tau}}, \quad (12)$$

which exceeds wage income $w_A L_A$ for any $\tau > 0$, reflecting the rebated tariff revenue. Equilibrium consumption demands in country A are then

$$C_{T_A A} = \frac{\alpha_{T_A} I_A}{P_{T_A}}, \quad (13)$$

$$C_{T_B A} = \frac{\alpha_{T_B} I_A}{(1 + \tau)eP_{T_B}}, \quad (14)$$

$$C_{N_A} = \frac{\alpha_N I_A}{P_{N_A}}. \quad (15)$$

In country B , which does not impose tariffs, income equals wage income and consumption demands remain

$$C_{TAB} = \frac{\alpha_{TA} w_B L_B}{P_{TA}/e}, \quad (16)$$

$$C_{TBB} = \frac{\alpha_{TB} w_B L_B}{P_{TB}}, \quad (17)$$

$$C_{NB} = \frac{\alpha_N w_B L_B}{P_{NB}}. \quad (18)$$

Tariffs do not directly affect production decisions in this environment. Firms operate under perfect competition and linear technologies, so equilibrium wages continue to satisfy $w_i = A_{T_i}$ and nontradable prices satisfy $P_{N_i} = A_{T_i}/A_{N_i}$, as in the free-trade case. The tariff alters only the consumer price of the imported good in country A and leaves producer prices unchanged in both countries, so sectoral labor allocations are also unchanged. As in the free-trade case, equilibrium is fully characterized by the trade balance condition $TB_A = 0$. Figure 1 illustrates the deviation from trade balance as a function of the exchange rate for different tariff rates. An increase in the tariff shifts the trade balance schedule upward at any given exchange rate, reflecting the reduction in import demand induced by the higher consumer price of the foreign tradable. Substituting (12)

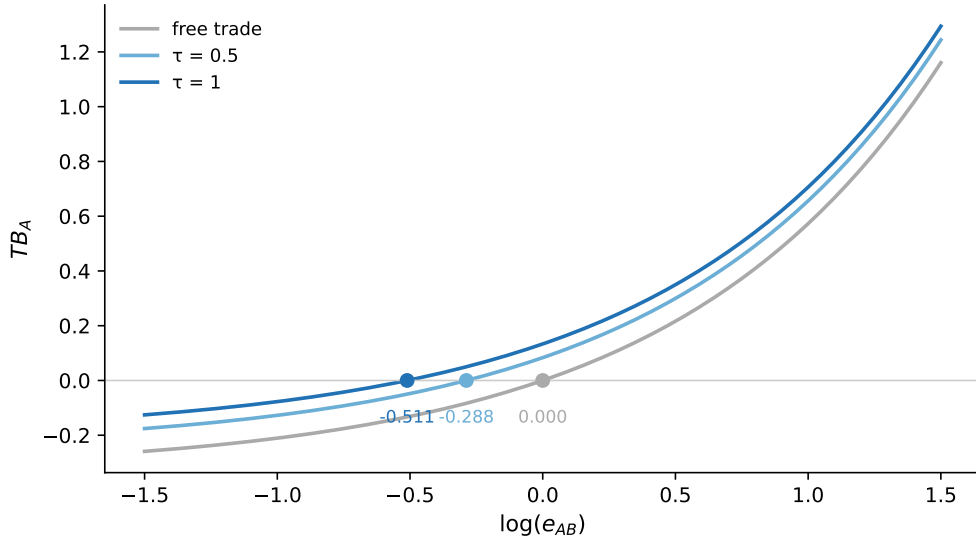


Figure 1: Trade balance deviation as a function of the exchange rate in the two-country model with tariffs. Equilibrium occurs where the trade balance deviation crosses zero. The figure is constructed under $A_{N_i} = A_{T_i} = 1$, $\alpha_{T_i} = \alpha_N = 1/3$, and the normalization $P_{TA} = P_{TB} = 1$. In this calibration, the free-trade equilibrium exchange rate is $e = 1$, and higher tariff rates lead to a larger equilibrium exchange rate appreciation.

and the demand expressions into the trade balance condition $TB_A = 0$ and solving for e yields

$$e(\tau) = \frac{1}{1+\tau} \cdot \frac{1}{1 - \frac{\alpha_{T_B} \tau}{1+\tau}} \cdot \frac{\alpha_{T_B}}{\alpha_{T_A}} \cdot \frac{A_{T_A} L_A}{A_{T_B} L_B}, \quad (19)$$

which is strictly decreasing in τ for $\alpha_{T_B} > 0$. An increase in tariffs by country A therefore requires an appreciation of its currency relative to country B in equilibrium in order to offset the tariff-induced improvement in the trade balance at a given exchange rate.

2.2 Robustness of the Two-Country Conventional Wisdom

The closed-form result in [Section 2.1](#) relies on Cobb–Douglas preferences and linear production technologies. We now show that the qualitative prediction $de/d\tau < 0$ is independent of these assumptions: it follows from general monotonicity properties of import and export behavior. [These conditions have familiar names: the exchange-rate condition of Lemma 2.2 is the Marshall–Lerner condition, and the tariff condition of Lemma 2.1 is its tariff analogue, that import expenditures fall as the consumer import price rises.](#) The summary of this subsection is therefore that the conventional appreciation result holds wherever Marshall–Lerner holds. This matters for [Section 3](#), where we show that [bilateral Marshall–Lerner conditions do not pin down the sign of the multilateral exchange-rate response](#). To establish this, we avoid solving each alternative specification in closed form and instead derive the sign of $de/d\tau$ from primitive conditions on trade flows. Equilibrium is characterized by balanced trade for country A ,

$$TB_A(e, \tau) \equiv X_A(e, \tau) - M_A(e, \tau) = 0,$$

where e denotes units of country A 's currency per unit of country B 's currency and τ is an ad valorem tariff imposed by country A on imports from country B . Assume that $TB_A(e, \tau)$ is continuously differentiable and that $TB_{A,e}(e, \tau) \neq 0$ locally. By the implicit function theorem, there exists a locally unique equilibrium mapping $e(\tau)$ satisfying

$$\frac{de(\tau)}{d\tau} = -\frac{TB_{A,\tau}(e(\tau), \tau)}{TB_{A,e}(e(\tau), \tau)}.$$

Thus, $de/d\tau < 0$ follows if $TB_{A,\tau}(e, \tau) > 0$ and $TB_{A,e}(e, \tau) > 0$ at equilibrium. Throughout, let $X_A(e, \tau)$ and $M_A(e, \tau)$ denote the values of exports and imports measured in country A 's currency. Let $p_M(e, \tau)$ denote the consumer price in country A of imports from country B , and let $p_X(e)$ denote the price of country A 's exports faced by consumers in country B , expressed in B 's currency. Under the producer-price normalization $P_{T_A} = P_{T_B} = 1$,

$$p_M(e, \tau) = (1 + \tau)e, \quad p_X(e) = \frac{1}{e}.$$

Lemma 2.1 (Tariff monotonicity). *Fix e . Suppose (i) the consumer import price $p_M(e, \tau)$ is*

increasing in τ , and (ii) import expenditures $M_A(e, \tau)$ are decreasing in the consumer import price. In addition, suppose that any induced effect of τ on export revenues satisfies

$$X_{A,\tau}(e, \tau) \geq 0, \quad \text{or the weaker condition} \quad X_{A,\tau}(e, \tau) > M_{A,\tau}(e, \tau),$$

which allows export revenues to fall as long as they fall by less than import expenditures. Then $TB_{A,\tau}(e, \tau) > 0$.

Proof. Differentiating $TB_A(e, \tau) = X_A(e, \tau) - M_A(e, \tau)$ with respect to τ at fixed e yields $TB_{A,\tau}(e, \tau) = X_{A,\tau}(e, \tau) - M_{A,\tau}(e, \tau)$. By (i)–(ii), $M_{A,\tau}(e, \tau) = (\partial M_A / \partial p_M)(\partial p_M / \partial \tau) < 0$. The stated condition on $X_{A,\tau}$ then implies $TB_{A,\tau}(e, \tau) > 0$. $\square$

Remark 1. In the baseline two-country model, the tariff applies only to imports and there is no policy response abroad, so $X_A(e, \tau)$ is independent of τ at fixed e . Hence $X_{A,\tau}(e, \tau) = 0$ and the condition of Lemma 2.1 holds trivially.

Lemma 2.2 (Exchange-rate monotonicity). *Fix τ . Suppose that a depreciation of country A's currency (an increase in e) weakly improves its trade balance, in the sense that*

$$M_{A,e}(e, \tau) < 0 \quad \text{and} \quad X_{A,e}(e, \tau) \geq 0,$$

with at least one inequality strict. Then $TB_{A,e}(e, \tau) > 0$.

Proof. Differentiating $TB_A(e, \tau) = X_A(e, \tau) - M_A(e, \tau)$ with respect to e at fixed τ yields $TB_{A,e}(e, \tau) = X_{A,e}(e, \tau) - M_{A,e}(e, \tau)$. Under the stated conditions, $-M_{A,e}(e, \tau) > 0$ and $X_{A,e}(e, \tau) \geq 0$, with at least one strict inequality, implying $TB_{A,e}(e, \tau) > 0$. $\square$

Proposition 2.1 (Sign of exchange rate response to tariffs). *Suppose that $TB_A(e, \tau)$ is continuously differentiable and that $TB_{A,e}(e, \tau) \neq 0$ in a neighborhood of an equilibrium (e^*, τ) satisfying $TB_A(e^*, \tau) = 0$. If the conditions of Lemmas 2.1 and 2.2 hold in that neighborhood, then the equilibrium mapping $e(\tau)$ exists locally and satisfies $de(\tau)/d\tau < 0$.*

Proof. By the implicit function theorem, there exists a locally unique function $e(\tau)$ such that $TB_A(e(\tau), \tau) = 0$, and $de(\tau)/d\tau = -TB_{A,\tau}(e(\tau), \tau)/TB_{A,e}(e(\tau), \tau)$. Lemma 2.1 implies $TB_{A,\tau}(e, \tau) > 0$ and Lemma 2.2 implies $TB_{A,e}(e, \tau) > 0$, so $de(\tau)/d\tau < 0$. $\square$

Lemmas 2.1 and 2.2 are satisfied under standard alternative preference, technology, and fiscal specifications because those alternatives preserve the Marshall–Lerner logic; the supporting arguments are deferred to Appendix A. The correct reading of this robustness is not that the appreciation result holds across all models, but that it holds wherever the bilateral Marshall–Lerner condition holds. The next section shows that in a three-country world this is no longer enough: every bilateral Marshall–Lerner condition can hold and the multilateral response can still go either way, depending on substitution patterns across foreign suppliers that the two-country model cannot express.

2.3 The Exchange Rate as a Relative Wage

Before moving to three countries, we state what the exchange rate is in this class of models. The model is real and has no nominal anchor: e is determined by a balanced-trade condition, not by a monetary equilibrium. Under the producer-price normalization $P_{T_i} = 1$ in each country's own currency, competitive pricing implies $w_i = A_{T_i}$ in own-currency units, so

$$\frac{\text{B's wage in A's currency}}{\text{A's wage}} = \frac{e A_{T_B}}{A_{T_A}} \propto e.$$

The nominal exchange rate is therefore proportional to the relative wage measured in a common currency, and because producer prices equal unit labor costs, it also equals the terms of trade. These objects coincide because there is one factor, constant returns, balanced trade, and no nominal rigidity. Statements below about depreciation of A 's currency are statements about A 's relative wage and terms of trade, interpreted as long-run outcomes once prices have adjusted.

This object is familiar in the trade literature. An isolated tariff by A on B , with τ_{AC} held at zero, is formally a preferential arrangement in favor of C : raising τ_{AB} and cutting τ_{AC} create the same discriminatory wedge, up to the level of the external tariff and the treatment of revenue. The third-country effects of such wedges are the subject of the preferential-trade-agreement literature, from Viner (1950) through Mundell (1964) on tariff preferences and the terms of trade, Kemp and Wan (1976), and the terms-of-trade externality framework of Bagwell and Staiger (1999) and Bagwell and Staiger (2002); Chang and Winters (2002) measure the price effects of Mercosur on excluded countries. That literature focuses on welfare, but the mechanism it works through is the bilateral terms of trade, which in the present model class is the exchange rate. On the quantitative side, Caliendo and Parro (2015), Ossa (2014), and Costinot and Rodríguez-Clare (2014) solve for these relative-wage responses to arbitrary tariff vectors in many-country settings, and Fajgelbaum et al. (2024) document the reallocation of trade toward untariffed origins empirically. What we add is: (i) a closed-form characterization of the sign of the third-country response to a discriminatory tariff, and of how it depends on the elasticity structure, home bias, country size, and tariff size; (ii) an impossibility result showing that with a single common elasticity the conventional two-country sign always survives, which we believe is why the point has been overlooked; and (iii) the translation into exchange-rate terms. The quantitative trade literature reports wage and welfare responses, but has not noted that its own equilibrium object implies that a purely real tariff shock can depreciate the dollar against the euro. That is the inference at issue in the 2025 debate.

2.4 Scope and Modelling Choices

Each restriction in the framework below is chosen for a reason, and we state the reasons here.

The long run. The paper characterizes the exchange rate consistent with a stationary external position, once prices have adjusted. This rules out nominal rigidities, which govern the speed of adjustment, and portfolio flows, which govern deviations during the transition. Neither changes

what the trade channel alone eventually requires. Balanced trade corresponds to zero net foreign assets in the intertemporal budget constraint. This normalization does affect the location of the threshold: solving the model with permanent trade imbalances $TB_i = -b_i$, a US-scale deficit of 3 percent of income raises the symmetric-point threshold from 3.96 to between 4.6 and 5.1, depending on the counterpart of the deficit, while a surplus lowers it. The structure of the result is unaffected: by the general result of Appendix B, reversal occurs if and only if ρ exceeds a threshold determined by baseline shares and incomes, at any external position. At 2025-scale tariffs, the tariff-size effect, which lowers the threshold, dominates the deficit effect: the combined threshold at $\tau = 1.45$ and a 3 percent deficit is about 3.5 for the real rate.

One factor. A single mobile factor is a separate choice, not a consequence of the long-run focus. It makes producer prices equal to unit labor costs, so that the terms of trade equal the relative wage exactly rather than approximately. This exactness is what justifies translating trade-theoretic results into exchange-rate terms. Section 3.4.1 relaxes this assumption with sector-specific factors and shows that the sign results are, if anything, strengthened.

No capital. Domestic capital accumulation affects transition dynamics and shifts steady states. These are second-order concerns for a long-run sign result, and they are studied in the dynamic quantitative literature (Eaton et al., 2016; Caliendo et al., 2019). An international financial account is a deeper departure: it does not shift the balanced-trade condition but replaces it with an intertemporal one. That class of model answers different questions; ours characterizes what trade flows alone pin down.

Three countries. The choice of country count involves a trade-off. Symmetric n -bloc models (Krugman, 1991; Bond and Syropoulos, 1996; Yi, 2000) keep closed forms by removing the heterogeneity the 2025 episode requires: one cannot ask what happens when one bystander is small and highly substitutable while another is large and diversified. Quantitative many-country models (Caliendo and Parro, 2015; Ossa, 2014) keep the heterogeneity but give up analytical results. Three countries is the smallest number at which discrimination is possible, since one needs a tariffed and an untariffed foreign partner, and the largest at which sign conditions remain available in closed form. The linearized system's numerator is a polynomial of degree $n - 1$ in ρ , so $n = 3$ gives a quadratic with a single relevant root, while at $n = 4$ a single threshold is no longer guaranteed.² The cost of the choice is that with a single untariffed aggregate, the model cannot show appreciation against some bystanders and depreciation against others at the same time, which is what the 2025 cross-section of bilateral rates shows. A natural extension would use the threshold of Proposition 3.2, in its general form, as a pair-by-pair diagnostic inside a calibrated many-country model.

²In four-country experiments we find asymmetric configurations in which the sign of $d \log e_{AC} / d\tau$ changes twice as ρ rises, so no single threshold exists.

3 The Three-Country Model

This section develops a three-country extension of the baseline framework and uses it to study how exchange rates adjust to tariffs when trade is multilateral. [The robustness result of Section 2.2 is useful here: every result below holds in a model in which each bilateral Marshall–Lerner condition is satisfied, so none of what follows reflects a failure of the two-country logic on its own terms. The question is what that logic leaves undetermined once a third country is present. The answer depends on a distinction between substitution margins that the two-country model cannot express.](#)

3.1 Model Setup

We develop a static three-country model with CES preferences over differentiated tradable goods and flexible exchange rates, which allows us to isolate the role of substitution patterns across foreign suppliers.

3.1.1 Firms

Each country $i \in \{A, B, C\}$ produces one distinct tradable good T_i and one non-tradable good N_i using labor as the sole factor of production. Technologies are linear:

$$Y_{T_i} = A_{T_i} L_{T_i}, \quad Y_{N_i} = A_{N_i} L_{N_i}, \quad L_{T_i} + L_{N_i} = L_i, \quad (20)$$

where A_{T_i} and A_{N_i} denote sectoral productivities and L_i is the labor endowment in country i . Markets are perfectly competitive. As in the two-country model, we normalize the producer price of each tradable good to one, $P_{T_i} = 1$, and maintain this normalization throughout. Under linear technologies and perfect competition, firms earn zero profits and wages are equalized across sectors within each country. Equilibrium wages therefore satisfy $w_i = A_{T_i}$, with nontradable prices pinned down by $P_{N_i} = A_{T_i}/A_{N_i}$, exactly as in the two-country model.

3.1.2 Households and Tariffs

Households in each country $i \in \{A, B, C\}$ consume all three tradable goods and a locally produced non-tradable good. Preferences are nested: households allocate income between a tradable bundle and a non-tradable good via Cobb–Douglas, and substitute across tradable varieties via a CES aggregator:

$$C_i = C_{T_i}^{\alpha_T} C_{N_i}^{\alpha_N}, \quad C_{T_i} = \left[\sum_{j \in \{A, B, C\}} \alpha_{T_j}^{1/\sigma} C_{T_j i}^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}},$$

where $\alpha_T = \alpha_{T_A} + \alpha_{T_B} + \alpha_{T_C}$, $\alpha_N = 1 - \alpha_T$, $\alpha_{T_j} > 0$, and $\sigma > 0$ is the elasticity of substitution across tradable varieties. The outer Cobb–Douglas layer fixes the expenditure share on the tradable bundle at α_T regardless of relative prices; the inner CES aggregator governs substitution across country varieties of tradables. Preference parameters are assumed to be identical across countries.

The Cobb–Douglas model of Section 2 arises as the special case $\sigma = 1$, under which the inner CES collapses to Cobb–Douglas across all four goods.

Let e_{ij} denote the nominal exchange rate defined as units of country i 's currency per unit of country j 's currency, with $e_{ii} = 1$ and the no-arbitrage condition $e_{ij} = e_{ik}e_{kj}$. In the three-country model the full exchange rate matrix is determined by two free bilateral rates, which we take to be e_{AB} and e_{AC} ; all other rates follow from no-arbitrage. Country i applies an ad valorem tariff $\tau_{ij} \geq 0$ to consumption of tradable good T_j . Under the producer-price normalization $P_{T_j} = 1$, the consumer price of T_j in country i is $(1 + \tau_{ij})e_{ij}$, with the tariff wedge $\tau_{ij}e_{ij}$ collected as government revenue and rebated lump-sum to households. The household budget constraint is

$$w_i L_i + T_i^{\text{rev}} = \sum_{j \in \{A, B, C\}} (1 + \tau_{ij}) e_{ij} C_{T_j i} + P_{N_i} C_{N_i}, \quad (21)$$

where $T_i^{\text{rev}} = \sum_j \tau_{ij} e_{ij} C_{T_j i}$. Letting $I_i = w_i L_i + T_i^{\text{rev}}$ denote disposable income, the outer Cobb–Douglas layer allocates a fixed share α_T of income to the tradable bundle, and the inner CES demand gives

$$C_{T_j i} = \frac{\alpha_{T_j} ((1 + \tau_{ij}) e_{ij})^{-\sigma}}{\sum_k \alpha_{T_k} ((1 + \tau_{ik}) e_{ik})^{1-\sigma}} \alpha_T I_i, \quad (22)$$

$$C_{N_i} = \frac{\alpha_N I_i}{P_{N_i}}. \quad (23)$$

With the utility weights entering as $\alpha_{T_j}^{1/\sigma}$, the weights α_{T_j} are themselves the expenditure shares at unit relative prices, which is the convention we maintain throughout and the one used in the calibration. The CES price index for the tradable bundle and the aggregate price level are

$$P_{T_i} = \left[\sum_{j \in \{A, B, C\}} \frac{\alpha_{T_j}}{\alpha_T} ((1 + \tau_{ij}) e_{ij})^{1-\sigma} \right]^{\frac{1}{1-\sigma}}, \quad P_i = P_{T_i}^{\alpha_T} P_{N_i}^{\alpha_N}, \quad (24)$$

and the bilateral real exchange rate between countries i and j is $q_{ij} \equiv e_{ij} P_j / P_i$. The outer Cobb–Douglas layer fixes total tradable expenditure at $\alpha_T I_i$, but for $\sigma \neq 1$ the division of that budget across varieties depends on relative prices. Let

$$s_{ij} \equiv \alpha_T \frac{\alpha_{T_j} ((1 + \tau_{ij}) e_{ij})^{1-\sigma}}{\sum_k \alpha_{T_k} ((1 + \tau_{ik}) e_{ik})^{1-\sigma}} \quad (25)$$

denote the realized share of total income that country i spends on variety j (inclusive of tariffs), with $\sum_j s_{ij} = \alpha_T$. Substituting the demand expressions into the tariff revenue formula and solving for I_i yields

$$I_i = \frac{w_i L_i}{1 - \sum_{j \in \{A, B, C\}} s_{ij} \frac{\tau_{ij}}{1 + \tau_{ij}}}, \quad (26)$$

which remains a closed form because the shares ς_{ij} depend only on prices, not on income. At $\sigma = 1$ the realized shares collapse to the constants $\varsigma_{ij} = \alpha_{T_j}$ and (26) reduces to the natural generalization of the two-country income formula (12); for $\sigma \neq 1$, using the raw weights α_{T_j} in place of ς_{ij} mismeasures tariff revenue.³ Under free trade all $\tau_{ij} = 0$ and $I_i = w_i L_i$, while positive tariffs raise disposable income above wage income through the rebated revenue.

3.1.3 Market Clearing and Equilibrium

For each country $i \in \{A, B, C\}$, non-tradable goods are produced and consumed domestically. Market clearing requires $Y_{N_i} = C_{N_i}$, which using $Y_{N_i} = A_{N_i} L_{N_i}$, $P_{N_i} = w_i/A_{N_i}$, and the Cobb–Douglas outer layer ($P_{N_i} C_{N_i} = \alpha_N I_i$) gives equilibrium labor allocations

$$\frac{L_{N_i}}{L_i} = \alpha_N \frac{I_i}{w_i L_i}, \quad \frac{L_{T_i}}{L_i} = 1 - \frac{L_{N_i}}{L_i}, \quad (27)$$

which equals α_N under free trade and exceeds it when rebated tariff revenue raises I_i above wage income. Each tradable variety T_i is produced only in country i and consumed in all countries. World market clearing for tradable T_i requires $Y_{T_i} = \sum_k C_{T_i k}$, which using $Y_{T_i} = A_{T_i} L_{T_i}$ and (22) gives

$$A_{T_i} L_{T_i} = \sum_{k \in \{A, B, C\}} \alpha_T I_k \frac{\alpha_{T_i} ((1 + \tau_{ki}) e_{ki})^{-\sigma}}{D_k}, \quad i \in \{A, B, C\}, \quad (28)$$

where $D_k = \sum_m \alpha_{T_m} ((1 + \tau_{km}) e_{km})^{1-\sigma}$ is the within-tradables CES denominator for country k ; the non-tradable good does not enter D_k because the outer Cobb–Douglas layer fixes its expenditure share at α_N . Note that I_k here is the full disposable income from (26), which already incorporates tariff revenue. Trade balances are measured in each country’s own currency. Exports of T_i from country i to country k are valued at the producer price in i ’s own currency, $P_{T_i} C_{T_i k} = C_{T_i k}$, while imports of T_j into country i are valued at $e_{ij} C_{T_j i}$, the producer-price value in i ’s currency (the tariff wedge is a domestic transfer, not a payment abroad). The trade balance for country i is therefore

$$\text{TB}_i = \sum_{k \neq i} C_{T_i k} - \sum_{j \neq i} e_{ij} C_{T_j i}, \quad i \in \{A, B, C\}. \quad (29)$$

An equilibrium is a pair of nominal exchange rates $\{e_{AB}, e_{AC}\}$ such that $\text{TB}_i = 0$ for all i . With three countries, only two of the three trade-balance conditions are independent by Walras’ Law, so the system (29) evaluated at $i \in \{B, C\}$ pins down the two free exchange rates, with $\text{TB}_A = 0$ following automatically. All remaining exchange rates are recovered via no-arbitrage: $e_{BA} = 1/e_{AB}$, $e_{CA} = 1/e_{AC}$, and $e_{BC} = e_{AC}/e_{AB}$.

3.1.4 Equilibrium Visualization

³This distinction matters quantitatively. An earlier version of this paper used the fixed-weight formula for all σ , which overstated rebated revenue on tariffed imports at high σ and generated a spurious sign reversal in e_{AC} . All results reported here use (25)–(26).

Panel (a) of Figure 2 illustrates equilibrium determination in a symmetric baseline calibration under Cobb-Douglas preferences ($\sigma = 1$), with $A_{T_i} = A_{N_i} = 1$ and $\alpha_{T_A} = \alpha_{T_B} = \alpha_{T_C} = \alpha_N = 1/4$. The panel plots, in (e_{AB}, e_{AC}) space, the loci of exchange rates that satisfy $TB_B = 0$ and $TB_C = 0$. By symmetry, the two loci are mirror images around the 45-degree line and intersect at $e_{AB} = e_{AC} = 1$. The bottom-left region corresponds to combinations of (e_{AB}, e_{AC}) for which both B and C run trade surpluses against A , reflecting an overvaluation of A 's currency relative to both foreign currencies; the top-right region corresponds to the reverse. The remaining two quadrants feature offsetting imbalances, with one foreign country in surplus and the other in deficit. Figure 2 collects the balanced-trade loci for the free-trade benchmark and for each of the three tariff experiments that follow in a single display.

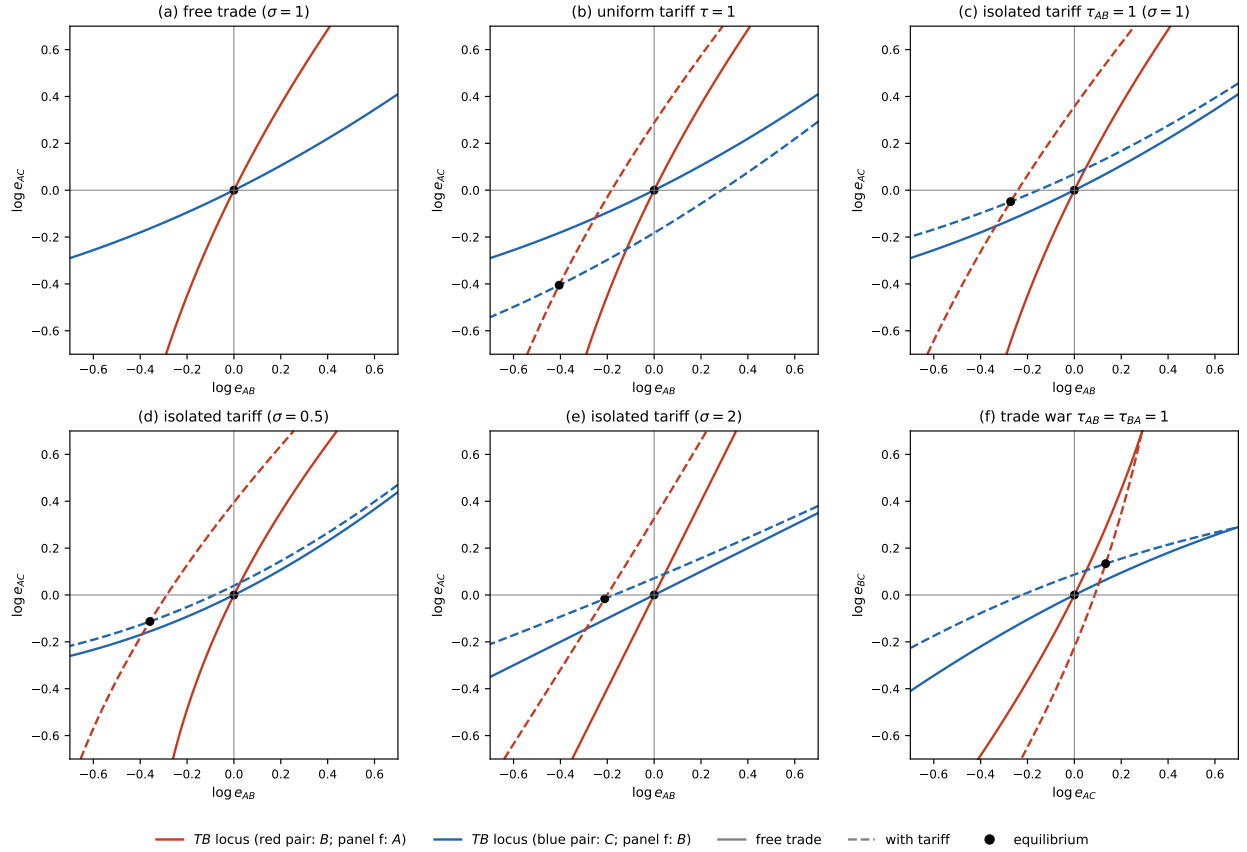


Figure 2: Balanced-trade loci and equilibria across the tariff experiments (symmetric baseline, $A_{T_i} = A_{N_i} = 1$, $\alpha_{T_i} = \alpha_N = 1/4$). Panels (a)–(e) plot the $TB_B = 0$ (red) and $TB_C = 0$ (blue) loci in $(\log e_{AB}, \log e_{AC})$ space; panel (f) plots the $TB_A = 0$ (red) and $TB_B = 0$ (blue) loci in $(\log e_{AC}, \log e_{BC})$ space. Solid: free trade; dashed: with the indicated tariff; dots: equilibria. (a) free trade; (b) uniform tariff $\tau = 1$; (c)–(e) isolated tariff $\tau_{AB} = 1$ at $\sigma = 1, 0.5, 2$; (f) trade war $\tau_{AB} = \tau_{BA} = 1$.

3.2 Tariff Experiments

This section studies the exchange rate response to tariffs in the three-country model. We consider three tariff configurations that progressively depart from the two-country benchmark: uniform tariffs, isolated tariffs, and bilateral trade war.

3.2.1 Uniform Tariffs: The Two-Country Analogue

We begin by considering a uniform tariff imposed by country A on imports from both foreign countries, B and C . Let $\tau \geq 0$ denote the common tariff applied to imports of both T_B and T_C , so that $\tau_{AB} = \tau_{AC} = \tau$ and all other tariff rates are zero. With linear technologies and competitive firms, producer prices remain normalized at $P_{T_i} = 1$, wages satisfy $w_i = A_{T_i}$, and nontradable prices $P_{N_i} = A_{T_i}/A_{N_i}$ are unchanged. The tariff applies only to consumer prices in country A and not to producer prices received abroad, so production decisions and sectoral labor allocations are unaffected. Panel (b) of Figure 2 plots the trade-balance loci for countries B and C under free trade ($\tau = 0$, solid lines) and a uniform tariff ($\tau = 1$, dashed lines), under the symmetric baseline calibration $A_{T_i} = A_{N_i} = 1$ and $\alpha_{T_i} = \alpha_N = 1/4$. Introducing a uniform tariff shifts both trade-balance loci outward relative to the 45-degree line, reflecting the reduction in import demand at any given exchange rate, and the new equilibrium corresponds to an appreciation of country A 's currency relative to both foreign currencies. Because the tariff is symmetric across foreign suppliers, country A 's currency appreciates by the same amount against e_{AB} and e_{AC} . Changes in the elasticity of substitution affect the magnitude of the appreciation but not its sign. This configuration therefore provides the natural three-country analogue to the two-country benchmark: uniform tariffs replicate the conventional result unambiguously.

3.2.2 Isolated Tariff: The Ambiguous Case

We next consider an isolated tariff imposed by country A on imports from country B only, so that $\tau_{AB} = \tau > 0$ and $\tau_{AC} = 0$. This breaks symmetry across foreign suppliers and introduces an adjustment margin absent in the uniform-tariff case. Panel (c) of Figure 2 plots the balanced-trade loci comparing free trade with an isolated tariff ($\tau_{AB} = 1$, dashed lines). The tariff shifts country B 's balanced-trade locus outward, reflecting reduced access to A 's market, while country C 's locus shifts inward. As a result, e_{AB} unambiguously falls — country A 's currency appreciates against B 's — whereas the sign of the response of e_{AC} is [not pinned down by the two-country logic](#) and depends on the relative magnitude of the two shifts. The ambiguity in e_{AC} reflects two opposing general-equilibrium forces. The first is a trade-diversion effect: imports into A are reallocated away from B toward C , raising demand for C 's exports and putting appreciation pressure on C 's currency. The second is a trade-redirection effect: as B loses access to A 's market, it reallocates exports toward C , increasing C 's imports and generating depreciation pressure on C 's currency. [A third force operates alongside these two: the tariff also switches \$A\$'s expenditure toward \$A\$'s own goods, the conventional two-country channel, which strengthens \$A\$'s trade position and pushes](#)

toward appreciation against both partners. Whether diversion can dominate is the central question of this section. Under a single elasticity, we show in Section 3.3 that it cannot.

When σ is low, trade diversion is muted and the redirection channel dominates, so the equilibrium remains close to the two-country benchmark and country A 's currency appreciates substantially against C as well as against B . Panel (d) of Figure 2 illustrates this case for $\sigma = 0.5$: at the symmetric baseline with $\tau_{AB} = 1$, $\log e_{AC} = -0.11$. When σ is higher, trade diversion toward country C becomes stronger and the appreciation of A against C shrinks: at $\sigma = 2$ (panel e) it has fallen to $\log e_{AC} = -0.016$. The pattern is monotone: as σ grows, $\log e_{AC}$ rises toward zero from below. One might expect that at some sufficiently high σ the sign flips and A depreciates against C . It does not. Section 3.3 shows that under a single elasticity the response approaches zero but never crosses it, at this calibration for any σ . Country A 's tariff on B generates an appreciation of A 's currency relative to B at every elasticity, consistent with conventional intuition. The new content of the multilateral setting lies in the behavior of e_{AC} . Extracting it requires more than raising a single elasticity: substitution between B 's and C 's goods and substitution between home and imported goods are different margins, and the single-elasticity CES ties them together by construction.

3.2.3 Bilateral Trade War: The Amplification Case

Finally, we consider a bilateral trade war in which countries A and B impose reciprocal tariffs on one another, so that $\tau_{AB} = \tau_{BA} = \tau > 0$, while country C remains untaxed. Relative to the isolated-tariff case, symmetry is restored between A and B , which are now jointly engaged in tariff escalation from the perspective of the uninvolved third country. Panel (f) of Figure 2 plots the balanced-trade loci for countries A and B in (e_{AC}, e_{BC}) space, comparing free trade (solid lines) with a bilateral trade war ($\tau_{AB} = \tau_{BA} = 1$, dashed lines). The imposition of reciprocal tariffs shifts both countries' balanced-trade loci outward in (e_{AC}, e_{BC}) space. When A and B are of equal size and impose equal tariffs, the bilateral exchange rate e_{AB} is unchanged by symmetry, and adjustment occurs entirely through a joint depreciation of both currencies relative to country C . Note that the stability of e_{AB} is a property of the exactly symmetric case, not a general result: with $\tau_{AB} \neq \tau_{BA}$ or unequal country sizes, e_{AB} moves, and in the calibration of Section 4 it does. The joint depreciation against C is robust for wars with similar tariff rates on both sides. It is not unconditional: when one country's tariff is far larger than the other's, the configuration approaches the isolated-tariff case and the heavily-tariffing country can appreciate mildly against C . At the 2025 rates ($\tau_{AB} = 1.45$, $\tau_{BA} = 1.25$) the war is well inside the joint-depreciation region.

A caution on magnitudes, which previews Section 3.3: in the single-elasticity specification, the size of the joint depreciation is *decreasing* in σ beyond low values, because a higher common elasticity also strengthens each country's substitution toward its own goods. The diversion force is governed by the cross-origin elasticity ρ of the nested specification introduced below, and the joint depreciation is increasing in ρ when the home-import margin is held fixed. Conflating the two margins reverses the comparative static.

The trade-war experiment implies that the exchange rate response depends on the configuration: appreciation against the tariffed partner under a unilateral tariff, and multilateral depreciation under retaliation. This is the pattern that Corsetti et al. (2025) document in high-frequency responses to the 2018–20 and 2025 tariff announcements: the dollar appreciates when US tariffs are unilateral and depreciates when trading partners retaliate. In this model the pattern arises from balanced-trade adjustment alone.

3.3 An Impossibility Result and a Reversal Threshold

The experiments above examined the sign of e_{AC} numerically. This subsection characterizes it analytically and contains the two central results of the paper. Throughout we work at the symmetric point (equal sizes, common preferences, equal weights on the two foreign varieties) and consider a marginal isolated tariff $d\tau$ imposed by A on B . General baselines are treated at the end of the subsection and in Appendix B. Let α_D denote the home share of tradable expenditure and, as before, α_T the tradable share of total expenditure.

Impossibility under a common elasticity. Consider first the flat specification of Section 3, in which a single elasticity σ governs substitution across all tradable varieties, with home expenditure share α_D within tradables.

Proposition 3.1 (Impossibility). *In the single-elasticity model, the local response of the third-country exchange rate to an isolated tariff satisfies*

$$\left. \frac{d \log e_{AC}}{d\tau} \right|_{\tau=0} = - \frac{\sigma (3\alpha_D - 1) + 3(1 - \alpha_D)(1 - \alpha_T)}{3 [3\alpha_D(\sigma - 1) + \sigma + 1]}.$$

If $\alpha_D \geq 1/3$, the numerator is positive for every $\sigma > 0$: country A 's currency appreciates against the untariffed third country at any elasticity. At the symmetric baseline calibration ($\alpha_{T_j} = \alpha_N = 1/4$, so $\alpha_D = 1/3$, $\alpha_T = 3/4$), the response is exactly $-1/(12\sigma)$, which approaches zero from below as $\sigma \rightarrow \infty$ but never crosses it.

The proof is a special case of Proposition 3.2 below, setting $\rho = \eta = \sigma$. The intuition is as follows. A higher σ strengthens trade diversion: A 's imports reallocate more elastically from B toward C . But the same σ also governs substitution between imports and home goods, so it strengthens conventional expenditure switching in the same proportion. The threshold that the diversion force must clear rises one-for-one with the force itself. For any realistic home bias ($\alpha_D \geq 1/3$; actual home shares of tradable expenditure are far higher) the diversion force never wins, at any σ . Figure 3 (panel a, dashed) shows the resulting asymptote. We verify numerically that the conclusion is not an artifact of the local approximation: for τ up to 1.5 and σ up to 50, $\log e_{AC} \leq 0$ at every point.⁴

⁴We believe this result also explains why the multilateral case has gone unnoticed: the natural first extension, adding a third country to a one-elasticity CES model, cannot produce the reversal.

Separating the margins. The single elasticity is doing double duty. To separate the two margins, replace the flat tradable aggregator with the nested structure

$$C_{Ti} = \left[\alpha_D^{1/\eta} C_{Ti}^{\frac{\eta-1}{\eta}} + (1 - \alpha_D)^{1/\eta} M_i^{\frac{\eta-1}{\eta}} \right]^{\frac{\eta}{\eta-1}}, \quad M_i = \left[\sum_{j \neq i} \beta_j^{1/\rho} C_{Tji}^{\frac{\rho-1}{\rho}} \right]^{\frac{\rho}{\rho-1}},$$

where η is the home-import (macro) Armington elasticity, ρ is the cross-origin (micro) elasticity across foreign suppliers, and $\beta_j = 1/2$ at the symmetric point. The outer Cobb–Douglas layer over tradables and non-tradables is unchanged, and the single-elasticity model is the special case $\rho = \eta = \sigma$. The distinction between the two elasticities is empirically grounded: Feenstra et al. (2018) find cross-origin elasticities substantially above home-import ones, Broda and Weinstein (2006) document dispersion at the product level, and Fajgelbaum et al. (2024) study the China–Vietnam reallocation case.

Proposition 3.2 (Reversal threshold). *In the nested model at the symmetric point,*

$$\left. \frac{d \log e_{AC}}{d\tau} \right|_{\tau=0} = \frac{\rho - \rho^*}{3[3\alpha_D(\eta - 1) + \rho + 1]}, \quad \boxed{\rho^* = 3[1 + \alpha_D(\eta - 1) - \alpha_T(1 - \alpha_D)]}$$

The denominator is positive whenever $\eta \geq 1$; this is the local stability condition for the balanced-trade equilibrium. Hence country A’s currency depreciates against the untariffed third country in response to a marginal discriminatory tariff on B if and only if $\rho > \rho^$.*

The comparative statics of the threshold are all interpretable. ρ^* is increasing in home bias α_D , since a more home-biased A has less import expenditure to divert ($\partial \rho^* / \partial \alpha_D = 3[(\eta - 1) + \alpha_T] > 0$); increasing in η , since stronger home-import switching raises the threshold; and decreasing in the tradable share α_T . Note that home bias and the elasticity separation play different roles. Realistic home bias disciplines magnitudes and modestly raises the threshold; the separation of the two elasticities is what makes the reversal possible at all. Correcting home bias alone under a common elasticity leaves the impossibility result in force. A note on language is also useful here. The statement “Chinese and Vietnamese goods are closer substitutes than American and Chinese goods” sounds like it requires different elasticities for different country pairs, but it is simply the ρ -versus- η distinction: the first comparison is foreign-versus-foreign, the second home-versus-foreign. No bilateral elasticity matrix is needed. At US-realistic parameters (home bias $\alpha_D = 0.80$, tradable share $\alpha_T = 0.40$, $\eta = 1.5$ in the range of Feenstra et al. (2018)), the threshold is $\rho^* = 3.96$. Table 1 reports the threshold across the relevant parameter range: ρ^* lies between roughly 2.5 and 5.5. Whether estimated cross-origin elasticities exceed the threshold depends on the episode. A US-scale trade deficit raises the threshold (Section 2.4), the real-rate measure raises it (below), and tariff size lowers it. For 2025-scale tariffs, the combined threshold with the real rate, a 3 percent deficit, and $\tau = 1.45$ is about 3.5, within the estimated range. For small tariffs at a US-scale deficit it is 5–6.5, at the upper end of the estimates. The reversal is therefore empirically plausible at standard parameter values, but it cannot appear in any specification that conflates the two margins.

Figure 3 summarizes both results.

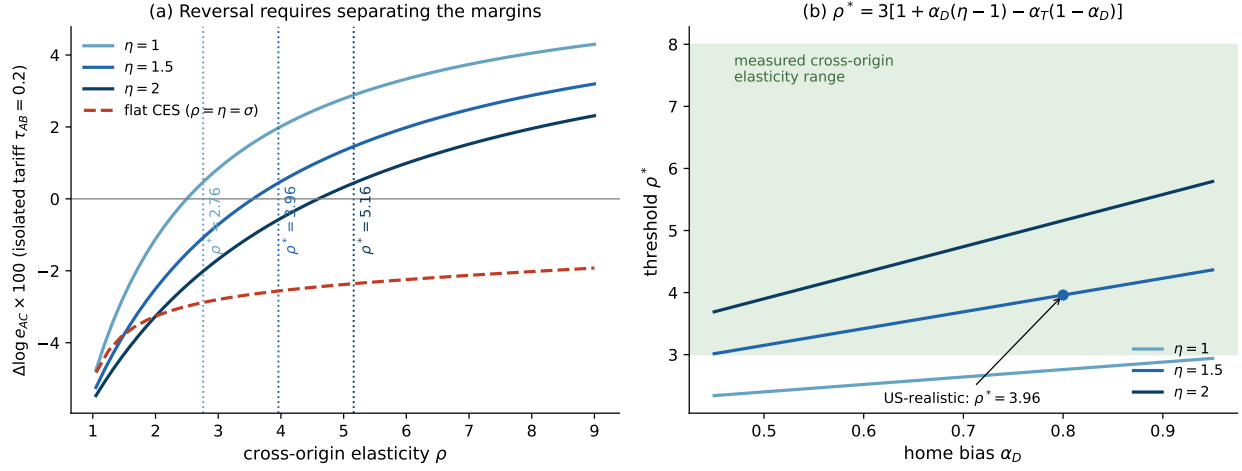


Figure 3: The reversal threshold. Panel (a): equilibrium change in $\log e_{AC}$ following an isolated tariff $\tau_{AB} = 0.2$, as a function of the cross-origin elasticity ρ , for three values of the macro elasticity η ($\alpha_D = 0.80$, $\alpha_T = 0.40$, equal sizes). Vertical dotted lines mark the analytic thresholds $\rho^*(\eta)$ of Proposition 3.2; each equilibrium path crosses zero slightly before its marked threshold because ρ^* is the marginal ($\tau \rightarrow 0$) threshold and the threshold falls with the tariff. The dashed path imposes $\rho = \eta = \sigma$ (the single-elasticity model): it approaches zero but never crosses it (Proposition 3.1). Panel (b): the threshold as a function of home bias α_D ; the shaded band indicates the range of measured cross-origin elasticities.

Third-country size. Panel B of Table 1 reports the threshold for unequal sizes, computed at the size configurations used in Section 4. The threshold is decreasing in the size of the third country: a small C cannot absorb diverted demand without a large wage response, so its currency appreciates sharply per unit of diverted expenditure, which pushes e_{AC} down. This matters for where one should expect to observe the reversal. Vietnam, often cited as the natural destination of trade diverted from China, faces by far the highest threshold ($\rho^* = 8.76$). Small, highly substitutable third countries are where the mechanism is hardest to see in bilateral rates, not easiest. Large aggregates such as the EU or the rest of the world, with thresholds near 4, are the natural place to look.

Large tariffs. Proposition 3.2 is local. Since the 2025 episode involves tariffs above 100 percent, Panel B also reports the discrete-tariff threshold: the value of ρ at which the cumulative change in $\log e_{AC}$ over $[0, \tau]$ equals zero. It falls monotonically in τ , from 3.96 at $\tau \rightarrow 0$ to 2.70 at $\tau = 1.45$ at equal sizes. The linearized threshold is therefore a conservative upper bound, and large tariffs enlarge the reversal region.

The real exchange rate. The propositions are stated for e , which is the relative wage and the terms of trade. The CPI-based real rate $q_{AC} = e_{AC}P_C/P_A$ is a distinct object because the tariff

Panel A: ρ^* by home bias and η ($\alpha_T = 0.40$)				Panel B: by third-country size / tariff		
	$\eta = 1.0$	$\eta = 1.5$	$\eta = 2.0$		L_C	ρ^*
$\alpha_D = 0.60$	2.52	3.42	4.32	Vietnam	0.055	8.76
$\alpha_D = 0.70$	2.64	3.69	4.74	EU	0.86	4.32
$\alpha_D = 0.80$	2.76	3.96	5.16	equal	1.00	4.24
$\alpha_D = 0.85$	2.82	4.09	5.37	ROW	3.51	3.78
$\alpha_D = 0.90$	2.88	4.23	5.58		τ	$\rho^*(\tau)$
					0.20	3.55
					0.50	3.19
					1.00	2.86
					1.45	2.70

Table 1: The reversal threshold ρ^* . Panel A: analytic threshold of Proposition 3.2 over the empirically relevant (α_D, η) box. Panel B, top: numerical threshold by third-country size at $\alpha_D = 0.80$, $\alpha_T = 0.40$, $\eta = 1.5$, $L_B = 1.21$ (sizes as in the Section 4 calibration). Panel B, bottom: the discrete-tariff threshold (the ρ at which the total change in $\log e_{AC}$ over $[0, \tau]$ is zero) at equal sizes; it falls monotonically with the tariff, so the local threshold is a conservative bound.

enters A 's consumer price index directly. Since $P_{N_i} = A_{T_i}/A_{N_i}$ is a productivity ratio unaffected by the tariff, all real-exchange-rate movement runs through tradables; the non-tradable sector plays no role in the mechanism but governs the mapping from e to q . This mapping matters for the threshold. The tariff raises A 's tariff-inclusive price index, which absorbs part of any nominal depreciation, so real depreciation against C requires more diversion than nominal depreciation does. At the benchmark parameters the real-rate threshold is $\rho_q^* = 4.93$, against $\rho^* = 3.96$ for the nominal rate. Both are inside the estimated cross-origin range. The calibration in Section 4 reports both objects.

General baselines. The symmetric-point formula generalizes. Linearizing the two balanced-trade conditions at an arbitrary free-trade baseline described by observables (common-currency incomes y_i , home shares h_i of tradable expenditure, and import-bundle origin shares b_{ij}), the local response takes the form $d \log e_{AC}/d\tau = N(\rho)/D$, where N is a quadratic in ρ with leading coefficient $b_{AB} b_{AC} b_{CA} b_{CB} (1 - h_A)(1 - h_C) y_A y_C > 0$. The threshold at any baseline is the larger root of N . The qualitative statement, that A depreciates against the untariffed third country if and only if ρ exceeds a threshold, therefore holds for arbitrary sizes and asymmetric trade patterns, with the threshold expressed in terms of baseline expenditure shares and incomes. Appendix B states the result and sketches the derivation; the symmetric point recovers Proposition 3.2 exactly.

3.4 Economic Mechanisms and Extensions

The tariff experiments illustrate that the standard two-country intuition for exchange-rate adjustment need not extend to multilateral trade environments. In a three-country setting, an isolated tariff imposed by country A on imports from B generates opposing trade-diversion and trade-redirection forces, and Section 3.3 showed exactly when diversion wins: never under a common

elasticity, and precisely when $\rho > \rho^*$ once the cross-origin and home–import margins are separated. The bilateral trade war case resolves this ambiguity: when both A and B impose reciprocal tariffs, the two forces reinforce one another from C ’s perspective, producing an unambiguous joint depreciation of A and B relative to C for near-symmetric wars.

The relative importance of the trade-diversion channel depends on several features that can be incorporated in richer models. First, as established above, the cross-origin elasticity ρ between goods produced by B and C in A ’s market is the central parameter, and it must be distinguished from the home–import elasticity η : raising a single common elasticity strengthens the diversion and expenditure-switching forces in lockstep and can never deliver the reversal (Proposition 3.1). Second, the strength of the diversion channel depends on expenditure patterns. When goods from country C account for a large share of A ’s import basket, relative price changes translate into larger demand reallocations toward C . Such asymmetries arise naturally from heterogeneous preference weights α_{T_j} across tradable varieties, and Appendix B shows how the threshold depends on them through baseline expenditure shares. Third, supply-side considerations matter, and here the exchange-rate implication is the opposite of the quantity intuition. Capacity constraints or increasing marginal costs in country C limit the quantity expansion of C ’s exports. But for that reason, diverted demand falls on C ’s price, which under the producer-price normalization is C ’s currency. Upward-sloping export supply therefore strengthens rather than dampens the appreciation of C per unit of diverted demand, and lowers the reversal threshold. We formalize this below with a sector-specific-factors extension.

Finally, initial trade patterns may also shape the strength of the trade-redirection channel. When the elasticity of demand in country C for goods from B is low, or when C ’s import capacity is limited, the redirection of B ’s exports following the tariff has a muted effect on C ’s import prices and hence on depreciation pressure on C ’s currency. This mechanism is related to the trade deflection literature (Bown and Crowley, 2007), who document empirically that import restrictions by one country redirect exports toward third market. Larger tariff shocks amplify the expenditure-switching and trade-diversion channels by inducing larger reallocations of trade flows (Amiti et al., 2019a).

3.4.1 Sector-Specific Factors: Upward-Sloping Export Supply

The linear technology makes export supply infinitely elastic at the going wage. To formalize the supply-side channel discussed above, we introduce a sector-specific fixed factor in tradables, as in the Ricardo–Viner model, replacing $Y_{T_i} = A_{T_i} L_{T_i}$ with

$$Y_{T_i} = A_{T_i} L_{T_i}^\gamma, \quad \gamma \leq 1,$$

with non-tradables linear as before and labor mobile across sectors within each country. Competitive pricing gives $w_i = \gamma A_{T_i} L_{T_i}^{\gamma-1}$ under the producer-price normalization, with the fixed factor earning rents $(1 - \gamma)P_{T_i} Y_{T_i}$ that are distributed lump-sum to households. The sectoral labor allocation now responds to the tariff, and $\gamma = 1$ recovers the baseline exactly. Equilibrium is a

five-equation system, non-tradable market clearing in each country plus two balanced-trade conditions, which we solve numerically.

There are two results, both in the direction suggested by the discussion above. First, at the symmetric benchmark the reversal threshold falls with the degree of diminishing returns, though modestly: ρ^* declines from 3.96 at $\gamma = 1$ to 3.83 at $\gamma = 0.5$. Second, and more important quantitatively, diminishing returns compress the size gradient of the threshold. Upward-sloping export supply binds most for a small third country, which cannot expand export quantities cheaply, so diverted demand is absorbed by its price. This is where the baseline mechanism was weakest:

	Vietnam ($L_C = 0.055$)	EU ($L_C = 0.86$)	ROW ($L_C = 3.51$)
$\gamma = 1$ (baseline)	8.76	4.32	3.78
$\gamma = 0.8$	6.07	4.18	3.78
$\gamma = 2/3$	5.27	4.09	3.77

Vietnam’s threshold falls by forty percent while the large aggregates barely move. Sector-specific factors therefore make small, supply-constrained third countries more plausible venues for the reversal, without changing any qualitative conclusion of Section 3.3. The converse also holds: none of the sign results require diminishing returns. The linear baseline delivers them, which is why we use it.

The boundary of the exercise. Two literatures mark where the static framework stops being the right tool. Dynamic quantitative trade models with capital accumulation (Eaton et al., 2016; Caliendo et al., 2019) are better suited for magnitudes and transition paths, since tariffs move investment demand and the path differs from the steady state. They are not needed for the sign mechanism, which requires neither capital nor dynamics. Intertemporal open-economy analysis (Obstfeld and Rogoff, 1996) adds a distinction the static model cannot express: a temporary tariff and a permanent tariff have different, and potentially opposite, exchange-rate implications once saving responds. Our static structure treats tariffs as permanent. This is a strong assumption for an episode marked by uncertainty about persistence (Kalemli-Özcan et al., 2025), and the results are best read as the long-run position conditional on the tariffs persisting.

4 Calibration Exercise

We calibrate the three-country model to offer quantitative guidance on the mechanisms identified in Section 3, using the 2025 US reciprocal tariff episode as the application. In this episode, the United States imposed large tariffs on imports from a range of trading partners, with particularly high rates directed at China and China retaliating in kind, culminating in effective bilateral tariff rates exceeding 100 percent by mid-2025. We consider three alternative configurations for the third economy C : the European Union, representing a large bloc of advanced-economy trading partners with deep trade links to both the US and China; Vietnam, representing a smaller economy that has emerged as a significant node for trade redirection away from the direct US-China bilateral linkage; and the rest of the world aggregated as a single block.

The model is deliberately stylized, and the calibration exercise should be interpreted not as a realistic simulation of the episode but as an illustration of the relative importance of the trade-diversion and trade-redirection channels under parameter configurations that are broadly consistent with its observable features. The parameters we vary across configurations are the expenditure shares $\{\alpha_{T_A}, \alpha_{T_B}, \alpha_{T_C}, \alpha_N\}$, which govern bilateral trade intensities; the labor endowments $\{L_A, L_B, L_C\}$, which govern relative economic size; and the tariff rates $\{\tau_{AB}, \tau_{BA}\}$ for the trade war experiment or τ_{AB} alone for the isolated tariff case. We set productivity parameters to $A_{T_i} = A_{N_i} = 1$ for all economies, so that relative economic size is captured entirely through the endowments L_i .

4.1 Tariff Regimes

We consider two tariff regimes corresponding to distinct phases of the 2025 US-China trade conflict, measured as incremental tariff rates above the pre-2025 baseline. The US is country A , China is country B , and the third country C varies across configurations. Table 2 summarizes the tariff rates across both regimes and configurations; full details of the rate construction are provided in Appendix C.

Regime 1 imposes a unilateral US tariff on China with no retaliation and no tariff on C , mapping directly onto the isolated tariff experiment of Section 3. Regime 2 represents peak bilateral escalation between the US and China; the rates are not exactly symmetric but are of similar magnitude and map naturally onto the trade war experiment. Furthermore, regime 2 includes $\tau_{AC} = 0.10$ across all configurations in this regime, reflecting the universal baseline tariff that remained in place for non-China trading partners after the April 9 pause. Regime 3 collects the rates in force at our December 2025 comparison window, after the May Geneva truce and the November 10 reduction of the fentanyl-related tariff: 20 percent on China (10 reciprocal plus 10 fentanyl-related), 10 percent Chinese retaliation, and the 2025 framework rates on third countries (15 percent for the EU, 20 percent for Vietnam, roughly 15 percent for the ROW aggregate). Throughout, each data window is compared with the tariffs actually in force in that window. This matters because the December structure is nearly uniform across China and third countries, which moves the model’s relevant experiment from the trade war toward the uniform-tariff case. Where announced-but-paused rates differed sharply from in-force rates (the April 2 schedule: 20 percent on the EU, 46 percent on Vietnam), we report an announced-rate variant in the replication package.

4.2 Preference and Endowment Calibration

The calibration uses the nested demand system of Section 3.3 with country-specific preferences calibrated from bilateral import shares. An earlier version of this paper used common preferences from world export shares, which implied, counterfactually, that the US spends more on Chinese tradables than on its own. For each country i in a configuration, the expenditure share on partner j ’s goods is measured as 2024 goods imports from j , importer-reported (US Census, China GACC, Eurostat, Vietnamese customs), divided by nominal GDP. The tradable expenditure share is $\alpha_T =$

Regime	Description	US-China		US- C		C -US	
		τ_{AB}	τ_{BA}	τ_{AC}^{EU}	τ_{AC}^{VNM}	τ_{CA}	Type
1	Fentanyl-related tariffs	0.20	0.00	0.00	0.00	0.00	Isolated
2	Peak escalation	1.45	1.25	0.10	0.10	0.00	Trade war
3	Late-2025 (post-truce)	0.20	0.10	0.15	0.20	0.00	Near-uniform

Table 2: Tariff regimes used in the calibration exercise. Rates are incremental ad valorem tariffs above the pre-2025 baseline. τ_{AB} is the US tariff on China, τ_{BA} is the Chinese tariff on the US, τ_{AC} is the US tariff on C , and τ_{CA} is C ’s tariff on the US. The ROW configuration uses the same τ_{AC} as EU and Vietnam.

0.40 throughout. The home tradable share α_{D_i} absorbs the residual, including goods from outside the trio, and the import-bundle weights β_{ij} follow from the partner split. Preference weights are then inverted at the model’s free-trade equilibrium: they are iterated until the equilibrium realized shares reproduce the data shares, since with unequal country sizes equilibrium prices differ from one and weights are not shares. Labor endowments remain proportional to 2024 PPP GDP (International Monetary Fund, 2024), and $A_{T_i} = A_{N_i} = 1$. Table 3 reports the implied home-bias parameters. US home bias is now 0.91–0.95 across configurations, in line with US spending on Chinese goods of 1–2 percent of expenditure.

A note on what “home” means in a three-country configuration. Within US tradable spending at border values, Chinese goods are 3.8 percent, EU goods 5.2 percent, and all foreign goods together 28 percent. The high home shares in the table (0.89–0.95) arise because goods from countries outside the trio must be assigned somewhere, and we assign them to home. Excluding non-trio imports from the basket instead gives $\alpha_{D_A} = 0.89$ and moves the results by a few tenths of a percentage point. The case with genuinely lower home bias, in which all foreign goods are treated as the composite partner and $\alpha_{D_A} = 0.72$, is the ROW configuration. The configurations therefore cover the possible treatments of non-trio trade, and the results are not sensitive to the choice.

Configuration	α_{D_A}	α_{D_B}	α_{D_C}	$s_{A,B}$	$s_{A,C}$	L_C/L_A
US–China–EU	0.911	0.941	0.881	1.50%	2.08%	0.86
US–China–Vietnam (adj.)	0.951	0.969	0.553	1.50%	0.47%	0.055
US–China–ROW	0.720	0.647	0.803	1.50%	9.69%	3.51

Table 3: Import-share calibration by configuration. α_{D_i} : country i ’s home share of tradable expenditure implied by 2024 bilateral goods imports over nominal GDP ($\alpha_T = 0.40$); $s_{A,j}$: US expenditure share on partner j ’s goods. The Vietnam configuration applies a processing-trade adjustment, treating half of Vietnam’s gross imports from China as final absorption (unadjusted, Vietnam’s gross imports from China are 30 percent of its GDP and its implied home share falls to 0.175); both variants are reported in the replication package. Sources and construction details in the replication package (`calibration_inputs.json`).

For the elasticities, we fix the home–import elasticity at $\eta = 1.5$, within the range of Feenstra et al. (2018), and report results over a grid of cross-origin elasticities $\rho \in [1.5, 8]$, with each configuration’s reversal thresholds marked, rather than assigning a single value per configuration. The

central empirical range for ρ is roughly 2–4. Median cross-origin estimates in Broda and Weinstein (2006) at three-digit aggregation lie near 2.5–4, and Li et al. (2022), who estimate the two-elasticity structure used here on bilateral trade, tariff, and production data, find the home–foreign elasticity 43 percent below the cross-foreign one, implying $\rho \approx 1.75\eta \approx 2.6$ at our η . Values of ρ can legitimately differ across configurations because each configuration is a self-contained three-country world: ρ measures the substitutability between that pair of suppliers at that level of aggregation, so the China–Vietnam elasticity may well exceed the China–ROW one (Fajgelbaum et al., 2024, 2020). This is now expressed by where each configuration sits on the reported grid rather than by an assigned point value.⁵

4.3 Results and Comparison With Data

Figure 4 reports model-implied Regime-2 exchange rate changes across the ρ grid alongside realized data. The data refer to percent changes⁶ in observed bilateral rates relative to the 2024 annual average, at two horizons: the April 2025 monthly average, compared with Regime 2, and the December 2025 monthly average, compared with Regime 3. Each window is matched to the tariffs in force in that window; comparing later data with the peak rates would confuse tariff changes with adjustment. Throughout, a positive change denotes a depreciation of the first-named currency: $\Delta e_{AC} > 0$ means the dollar depreciates against C ’s currency. For the ROW configuration we construct a matched comparison index. The model’s C is the world excluding the US and China, so we strip the renminbi component out of the Federal Reserve H.10 broad nominal dollar index using the Fed’s published weight (10.897 percent) and renormalize: $\Delta \log I^{\text{ex-CN}} = (\Delta \log I^{\text{broad}} - w_{\text{CN}} \Delta \log e_{\text{CNY/USD}})/(1 - w_{\text{CN}})$. This replaces the advanced-economy index used in an earlier version, which excluded emerging markets from the aggregate the model refers to. The replacement matters: the matched index shows the dollar appreciating slightly against the ex-China world in April 2025 (−0.95 percent), where the advanced-economy index showed depreciation.

April 2025 against Regime 2. Consider first the EU configuration, the cleanest available test of the bystander mechanism: the euro floats, the EU faced only the 10 percent baseline in April, and it did not retaliate. At $\rho = 2.0$, the model’s Regime-2 prediction is $(\Delta e_{AB}, \Delta e_{AC}) = (-0.80, +3.45)$ percent, against April 2025 data of $(-1.51, +3.36)$. The model matches both signs and both magnitudes at a cross-origin elasticity in the middle of the empirical range. The ROW configuration, evaluated against the matched ex-China index, is consistent at a somewhat higher elasticity. At $\rho = 3.0$ the model predicts $(-3.78, -1.34)$ against April data of $(-1.51, -0.95)$. In April the dollar appreciated slightly against the ex-China world, and the model reproduces this: at moderate ρ , the asymmetric war plus the 10 percent tariff on C itself leaves the dollar mildly strong against a large bystander.

⁵Appendix D retains six additional third-country configurations under the previous calibration for reference; their rebuild under the import-share approach requires additional bilateral data collection and is left to the replication package.

⁶We report $100 \times (\text{level}_t / \text{level}_0 - 1)$ relative to the 2024 annual average.

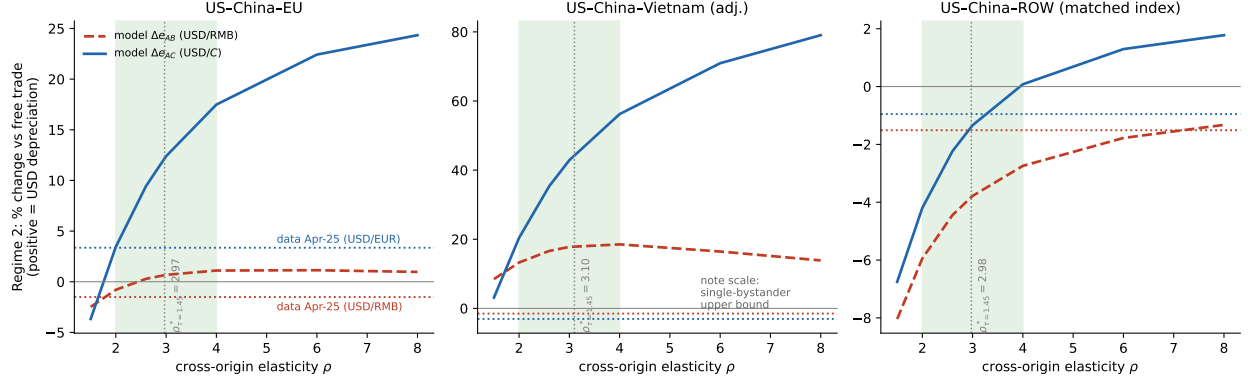


Figure 4: Regime-2 model predictions across the cross-origin elasticity grid, against the April 2025 data. The December window is compared with Regime 3 in the text. Solid: model Δe_{AC} (USD against C 's currency); dashed: model Δe_{AB} (USD/RMB); horizontal dotted lines: observed April 2025 changes relative to the 2024 average, colored to match the corresponding model object. Shaded band: the central cross-origin range $\rho \in [2, 4]$. Vertical dotted line: the configuration's cumulative reversal threshold at $\tau = 1.45$. Positive values denote dollar depreciation. The Vietnam panel is an upper bound (see text).

Vietnam is not a bystander test. The dollar strengthened against the dong throughout 2025 (−3.06 percent in April, −4.83 by December), while the trade-war experiment predicts the opposite. No defensible elasticity repairs this within the bystander framing: even using the announced-but-paused 46 percent April rate on Vietnam, the model has the dong appreciating, because the 145 percent rate on China still dominates. The resolution is that Vietnam was not a bystander. It was itself a principal tariff target, with 46 percent announced on April 2 and 20 percent in force from July, equal to the December rate on China. Once Vietnam's actual tariff is used (Regime 3), the model predicts dollar strength against the dong of 3.5–4.6 percent at central elasticities, against 4.8 percent observed. With near-equal rates on China and Vietnam there is nothing to divert between them, the configuration reduces to the conventional own-tariff case, and the small tariffed country's currency falls. The dong is therefore a test of the classification, in the sense that the model tracks who is tariffed and by how much, rather than of the bystander mechanism, for which the euro and the ex-China aggregate are the valid tests. The Vietnam configuration otherwise remains a mechanism illustration: with a single bystander one-eighteenth the size of the US, the model routes all of the diversion through one small economy, which overstates the response by construction.

December 2025 against Regime 3. By December the tariff structure was nearly uniform: 20 percent on China and 15–20 percent on third countries. The discrimination that drives the diversion mechanism had largely disappeared, and the model's prediction reverts toward the conventional uniform-tariff case of mild dollar appreciation against everyone. At $\rho = 2.6$ the predictions are: EU configuration (−4.6, −4.9), ROW (−5.6, −6.4), Vietnam (−4.6, −4.6). In the data, the dong matches (−4.8), but the euro (+8.2) and the ex-China index (+2.3) show substantial dollar weakness. The conclusion is that at December's near-uniform rates, the tariffs then in force cannot

account for the observed multilateral dollar depreciation through the trade channel. Whatever drove the late-2025 dollar, whether expectations that escalation would resume, an erosion of the dollar’s safe-haven status (Kalemli-Özcan et al., 2025; Cardani et al., 2025), or monetary policy, it was not the prevailing tariffs. The framework that attributes the April pattern to trade fundamentals thus declines to attribute the December pattern to them.

Three further results are worth noting. First, the model’s real-rate predictions track the nominal ones with a modest gap ($\Delta q_{AC} = +7.6$ against $\Delta e_{AC} = +9.4$ percent in the EU configuration at $\rho = 2.6$), consistent with the result in Section 3.3 that the real-rate reversal threshold exceeds the nominal one. Second, each configuration’s cumulative reversal threshold at the peak bilateral tariff is nearly the same: 2.97 (EU), 3.10 (Vietnam, adjusted), and 2.98 (ROW). At 2025-scale tariffs the reversal therefore requires a cross-origin elasticity near 3 in every configuration, within the estimated range. Third, under Regime 1 (the isolated tariff) the model concentrates appreciation on the renminbi, with a much smaller move against C : $\Delta e_{AB} = -5.6$ and $\Delta e_{AC} = -1.6$ percent in the EU configuration at $\rho = 2.6$, against March data of -0.91 and -0.23 . The model overpredicts the one-month bilateral magnitude, as a long-run model should at a one-month horizon.

Two caveats apply. First, monthly exchange-rate observations are being compared with a static balanced-trade model whose predictions are long-run tendencies. Exchange-rate disconnect at short horizons (Itskhoki and Mukhin, 2021) cautions against reading any single window as a point test, and the December window shows how much of the exchange rate the trade channel does not explain once the tariff structure turns uniform. Second, the three-country design limits what any configuration can capture. A single untariffed aggregate cannot show dollar appreciation against some bystanders and depreciation against others at the same time, which is what the cross-section of bilateral rates shows, so each configuration should be read as the model’s answer for that bystander in isolation. We therefore present the exercise as an illustration of the mechanism, with magnitudes on the scale of the data at central elasticities, rather than as a validated forecast.

5 Conclusion

This paper connects the tariff–exchange-rate question to the literature on preferential trade agreements. In the model class behind the conventional tariff–appreciation prediction, the exchange rate equals the relative wage and the bilateral terms of trade, which is the object that literature has studied since Viner, and a selective tariff is formally a preference in favor of the untariffed partner. We argue the connection has gone unstated because it is invisible in the standard specification. Under a single elasticity of substitution across all tradable varieties, the conventional prediction survives the move to three countries: a discriminatory tariff cannot depreciate the tariff-imposing country’s currency against the untariffed third country, at any elasticity, under realistic home bias, because a common elasticity strengthens trade diversion and home–import substitution in the same proportion. The reversal exists only once the two margins are separated. The currency then depreciates against untariffed partners if and only if the cross-origin elasticity exceeds the threshold $\rho^* = 3[1 + \alpha_D(\eta - 1) - \alpha_T(1 - \alpha_D)]$, which is about 4 at US-realistic parameters for small tariffs and

about 3 at 2025-scale tariffs, within the range of estimated cross-origin elasticities. In a bilateral trade war with similar rates on both sides, both currencies depreciate against the untariffed third country, with magnitude increasing in ρ . A calibration to the 2025 episode, in which each month of data is compared with the tariffs in force in that month, matches the April pattern of modest dollar appreciation against the renminbi and a 3–4 percent depreciation against the euro at a cross-origin elasticity of about 2. It also explains the dollar’s strength against the dong once Vietnam’s own tariff is used: the model tracks who is tariffed and by how much.

These results have direct implications for interpreting exchange-rate movements during tariff episodes, and the implications run in both directions. Exchange-rate appreciation against a bilateral trading partner is not evidence of broad currency strength, and observed multilateral depreciation during a discriminatory tariff episode cannot be attributed to financial channels on the basis of the two-country prediction alone. Real trade channels generate this pattern once cross-origin substitution exceeds the threshold, and generate it unambiguously in a retaliated trade war; this dependence on the configuration is documented in high-frequency data by Corsetti et al. (2025). The same framework also indicates when the attribution fails. By late 2025, with near-uniform rates across China and third countries, the trade channel predicts mild dollar strength, so the year-end dollar weakness must be attributed elsewhere, whether to expectations of renewed escalation, an erosion of safe-haven status, or monetary policy. The framework thus gives conditions under which the trade channel does or does not explain the data, which is what the attribution debate has lacked.

Several directions for future work suggest themselves. A natural extension would embed the mechanisms identified here in a quantitative multi-country trade model with richer bilateral trade flow patterns, input-output linkages, and heterogeneous elasticities across sectors and country pairs. A further extension would introduce nominal rigidities and a financial account, allowing the real exchange rate channels to interact with the monetary and portfolio balance dynamics that dominate exchange rate movements at shorter horizons. Incorporating these features into a unified framework would bring the conceptual results established here closer to a quantitative tool suitable for policy analysis.

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A Two-Country Conventional Wisdom Robustness

Alternative demand-side specifications. We first examine the robustness of Lemmas 2.1 and 2.2 under a broad class of standard demand systems. The key observation is that tariffs and exchange rates affect demand only through relative prices. As long as import demand is decreasing in its consumer price and export revenues are weakly increasing in export competitiveness, the monotonicity conditions required for the sign of $de/d\tau$ are satisfied. Consider CES (Armington) preferences over tradable goods with elasticity of substitution $\sigma > 0$, with utility separable between tradables and non-tradables. Under the producer-price normalization, the consumer price of imports in country A is $p_M(e, \tau) = (1 + \tau)e$. For any $\sigma > 0$, Marshallian import demand is strictly decreasing in $p_M(e, \tau)$, so at a fixed exchange rate an increase in τ reduces import expenditures while export revenues are locally unaffected, verifying the conditions of Lemma 2.1. Holding τ fixed, a depreciation raises the domestic-currency price of imports and lowers the foreign-currency price of exports, implying falling import expenditures and weakly rising export revenues, which verifies Lemma 2.2. The same logic applies to nested CES preferences that allow for separate elasticities between tradables and non-tradables and across tradable varieties. While tariff-induced income effects may reallocate expenditure between tradables and non-tradables, the monotonic relationship between import demand and its consumer price is preserved as long as the elasticity of substitution across tradables is positive, and exchange-rate movements continue to generate standard expenditure-switching effects. Finally, under quasi-linear preferences, income effects from tariff revenue are eliminated entirely. Import demand depends only on relative prices and remains decreasing in the consumer import price, while export revenues respond monotonically to changes in export competitiveness. The conditions of Lemmas 2.1 and 2.2 therefore hold by construction.

Alternative production-side specifications. We next examine alternative production structures that allow wages and output to adjust endogenously. The common feature of these specifications is that tariffs and exchange rates affect production only through their effects on demand and relative prices. Consider production technologies with diminishing returns to labor or sector-specific factors. Under perfect competition, wages adjust to clear labor markets and output responds endogenously to demand shifts. Holding the exchange rate fixed, a higher tariff continues to raise the consumer price of imports in country A and reduce import expenditures. Although wages and output adjust in general equilibrium, these adjustments do not overturn the import-compression effect of the tariff. In particular, since the tariff does not introduce any export-side wedge or subsidy, export revenues are not adversely affected at fixed e , so the conditions of Lemma 2.1 remain satisfied. Holding the tariff fixed, a depreciation raises the domestic-currency price of imports and lowers the foreign-currency price of exports. Under standard regularity conditions, import expenditures fall and export revenues weakly increase, even when wages and output adjust endogenously. Sectoral wage adjustments induced by demand shifts reinforce, rather than offset, the expenditure-switching effects as long as higher export demand raises export-sector wages or output without generating a sufficiently strong increase in marginal costs that would reverse the competitiveness gain. We rule

out such counterfactual cases by assuming that export supply is weakly increasing in export prices, under which the requirements of Lemma 2.2 are satisfied. A similar argument applies when export supply is endogenous due to capacity constraints or increasing marginal costs: provided that export supply is weakly increasing in the export price faced by foreign consumers, a depreciation raises the value of exports measured in country A 's currency while higher import prices reduce import expenditures, so $TB_{A,e}(e, \tau) > 0$ continues to hold locally.

Alternative fiscal closures and other specifications. Finally, we briefly examine additional modeling choices commonly considered in the literature. These specifications differ in how income effects, pricing, or market structure are treated, but they leave unchanged the relative-price channels through which tariffs and exchange rates affect trade flows in the medium to long run. First, consider alternative treatments of tariff revenue. Whether tariff revenue is rebated lump-sum, wasted, transferred abroad, or used to finance government spending, a tariff continues to raise the consumer price of imports at a fixed exchange rate. As long as import demand is decreasing in its consumer price, import expenditures fall at the prevailing exchange rate while export revenues are not adversely affected locally, and a depreciation continues to raise import prices and lower export prices. Hence the conditions of Lemmas 2.1 and 2.2 hold under standard fiscal closures. Second, consider alternative market structures and pricing assumptions, such as monopolistic competition in tradables or local- and dominant-currency pricing. These assumptions primarily affect the degree and timing of exchange rate passthrough. In the short run, incomplete passthrough under dominant-currency pricing may attenuate expenditure-switching effects and delay the adjustment of trade flows. However, once prices adjust, relative prices faced by consumers reflect both the tariff wedge and the exchange rate, and in this flexible-price equilibrium the monotonicity conditions remain valid. Finally, these conclusions are unchanged in simpler or asymmetric environments, including pure-tradables economies or the small open economy limit. In each case, tariffs compress import demand at a fixed exchange rate and exchange-rate movements generate standard expenditure-switching effects. While the magnitude of the exchange rate response may vary across specifications, its sign does not. Taken together, the results in this section show that the appreciation of the tariff-imposing country's currency in a two-country setting is not an artifact of particular assumptions about fiscal policy, market structure, or pricing conventions. Rather, it follows from general monotonicity properties of import and export behavior that hold across a wide range of standard models, once relative prices are allowed to adjust.

B The Reversal Threshold at a General Baseline

This appendix derives the local response $d \log e_{AC} / d\tau|_{\tau=0}$ at an arbitrary free-trade baseline of the nested model, by linearization in baseline expenditure shares ("hat algebra"). The baseline is described entirely by observables: common-currency incomes $y_i = e_{Ai} w_i L_i$, home shares h_i of each country's tradable expenditure, import-bundle origin shares b_{ij} (the share of origin j in i 's import bundle, $\sum_{j \neq i} b_{ij} = 1$), and the tradable expenditure share α_T .

Let $\nu_j = \log e_{Aj}$ (with $\nu_A = 0$) and let s_{ij} denote the share of country i 's total income spent on variety j , so $s_{ii} = \alpha_T h_i$ and $s_{ij} = \alpha_T(1 - h_i)b_{ij}$ for $j \neq i$. Log-differentiating the nested CES demand system at the baseline, with $d \log p_{ij} = d\nu_j + \mathbf{1}\{(i, j) = (A, B)\} d\tau$:

$$\begin{aligned} d \log P_{M_i} &= \sum_{j \neq i} b_{ij} d \log p_{ij}, & d \log P_{T_i} &= h_i d \log p_{ii} + (1 - h_i) d \log P_{M_i}, \\ d \log s_{ii} &= (1 - \eta) (d \log p_{ii} - d \log P_{T_i}), \\ d \log s_{ij} &= (1 - \rho) (d \log p_{ij} - d \log P_{M_i}) + (1 - \eta) (d \log P_{M_i} - d \log P_{T_i}), \quad j \neq i, \end{aligned}$$

and $d \log I_i = d\nu_i + \mathbf{1}\{i = A\} s_{AB} d\tau$, where the second term is the first-order tariff-rebate effect. Totally differentiating the two independent balanced-trade conditions

$$\text{TB}_i = \sum_{k \neq i} \frac{s_{ki} I_k}{1 + \tau_{ki}} - \sum_{j \neq i} \frac{s_{ij} I_i}{1 + \tau_{ij}}, \quad i \in \{B, C\},$$

at $\tau = 0$ (the term $-s_{AB} y_A d\tau$ from $d[1/(1 + \tau_{AB})]$ enters B 's export revenue) yields a linear 2×2 system in $(d\nu_B, d\nu_C)$. Solving by Cramer's rule:

Proposition B.1 (General-baseline threshold). *At any free-trade baseline,*

$$\left. \frac{d \log e_{AC}}{d\tau} \right|_{\tau=0} = \frac{N(\rho)}{D}, \quad N(\rho) = c_2 \rho^2 + c_1 \rho + c_0,$$

where $D > 0$ at a locally stable equilibrium and the coefficients c_2, c_1, c_0 depend only on $(y_i, h_i, b_{ij}, \alpha_T, \eta)$. The leading coefficient is

$$c_2 = b_{AB} b_{AC} b_{CA} b_{CB} (1 - h_A)(1 - h_C) y_A y_C > 0,$$

so N is an upward parabola and the reversal threshold is its larger root ρ^* : country A depreciates against the untariffed third country if and only if $\rho > \rho^*$, at any baseline. All three coefficients share the factor $b_{AB}(1 - h_A) y_A$, country A 's import exposure to B , and c_0 contains the factor $1 + h_A(\eta - 1) - \alpha_T(1 - h_A)$, which is the symmetric threshold $\rho^*/3$ with h_A in place of α_D .

At the symmetric point ($h_i = \alpha_D$, $b_{ij} = 1/2$, $y_i = 1$) the expression reduces exactly to Proposition 3.2. The full coefficient expressions are long, and are provided in the replication package, together with a symbolic derivation and a numerical validation against nonlinear equilibrium solves (agreement to five decimal places at the size configurations of Table 1).

C Calibration Details

Tariff Rate Calibration

All tariff rates are measured as incremental ad valorem rates above the pre-2025 baseline, which we treat as free trade. Rates are sourced from official executive orders, Federal Register notices,

and contemporaneous policy documentation.

Regime 1 (Fentanyl-related tariffs). The 20 percent tariff on Chinese imports ($\tau_{AB} = 0.20$) reflects the cumulative incremental tariffs imposed on China in February and March 2025 under the International Emergency Economic Powers Act (IEEPA) on fentanyl-related grounds, prior to the Liberation Day announcement. China did not impose retaliatory tariffs during this period, giving $\tau_{BA} = 0$. No incremental tariff was imposed on third countries at this stage, so $\tau_{AC} = \tau_{CA} = 0$ across all configurations.

Regime 2 (Peak escalation). Following the Liberation Day announcement of April 2, 2025, the US imposed country-specific reciprocal tariffs on most trading partners effective April 9. However, on April 9 the administration suspended all country-specific reciprocal tariffs for 90 days for every country except China, leaving only a universal 10 percent baseline tariff in place for non-China partners. The effective incremental US tariff on China at the peak of escalation was 145 percent ($\tau_{AB} = 1.45$), combining the Liberation Day reciprocal rate of 34 percent with the pre-existing fentanyl tariffs and subsequent escalation steps. China retaliated with a cumulative tariff of 125 percent on US goods ($\tau_{BA} = 1.25$). For all third-country configurations, the US tariff on C is set to the 10 percent baseline that remained in effect after the April 9 pause ($\tau_{AC} = 0.10$). Neither the EU nor Vietnam imposed retaliatory tariffs on the US during this period, so $\tau_{CA} = 0$.

Regime 3 (Late 2025). The May 12 Geneva agreement suspended the peak bilateral rates: the US rate on China fell to 30 percent incremental (10 percent reciprocal plus 20 percent fentanyl-related), and China’s retaliation to 10 percent. On November 10, 2025, following the October–November agreement, the fentanyl-related rate was halved, leaving the US rate on China at 20 percent ($\tau_{AB} = 0.20$) and China’s at 10 percent ($\tau_{BA} = 0.10$) through the December window. Third-country rates reflect the 2025 framework agreements: 15 percent on EU goods (July framework), 20 percent on Vietnamese goods (July agreement), and approximately 15 percent for the ROW aggregate (a rough trade-weighted average of the 10 percent baseline, the 15 percent floor for surplus countries, and higher country-specific rates). No retaliation by the EU or Vietnam ($\tau_{CA} = 0$).

Expenditure Shares

Expenditure shares are calibrated from 2024 world goods export shares (WTO), with the nontradable share fixed at $\alpha_N = 0.60$ to match the broad cross-country average. Letting s_j denote country j ’s share of world goods exports, tradable shares are constructed as

$$\alpha_{T_j} = 0.40 \times \frac{s_j}{s_A + s_B + s_C},$$

normalizing within each configuration so that shares sum to one. The underlying export shares are $s_A = 8.4\%$ (US), $s_B = 14.3\%$ (China), $s_C = 12.0\%$ (EU), 1.6% (Vietnam), and 77.2% (ROW),

where the ROW share is computed as world exports minus the US and China. Because the normalization denominator depends on s_C , the values of α_{T_A} and α_{T_B} differ across configurations; the resulting calibration targets are reported in Table 3. Preference parameters are assumed common across countries, so the same shares apply to A , B , and C .

Labor Endowments

Labor endowments are set proportional to 2024 GDP at purchasing power parity (PPP), normalizing $L_A = 1$ for the US. PPP-based weights are appropriate because the model abstracts from price level differences across countries, and PPP GDP provides a better measure of relative economic size in volume terms than nominal GDP at market exchange rates. The source figures from the IMF World Economic Outlook ([International Monetary Fund, 2024](#)) are: United States \$29.2 trillion, China \$35.3 trillion, European Union \$25.1 trillion, and Vietnam \$1.6 trillion (all in 2024 international dollars), giving $L_B = 1.21$, $L_C^{\text{EU}} = 0.86$, and $L_C^{\text{VNM}} = 0.055$. For the ROW configuration, L_C^{ROW} is computed as world PPP GDP minus the US and China, giving $\$167.0 - \$29.2 - \$35.3 = \102.5 trillion and $L_C^{\text{ROW}} = 3.51$.

Elasticity of Substitution

The elasticity of substitution within the tradable bundle is set to $\sigma = 6$ for the EU configuration, $\sigma = 8$ for Vietnam, and $\sigma = 2$ for the ROW configuration. For the EU, the value $\sigma = 6$ is anchored on the cross-origin central estimate from [Amiti et al. \(2019b\)](#), which yields a preferred elasticity near six across a broad range of manufactured goods imported by the US. The EU value reflects the China Shock 2.0 narrative, under which Chinese manufacturing increasingly competes with European industrial and machinery exports in US import markets. For Vietnam, $\sigma = 8$ reflects Vietnam’s deep integration into Chinese supply chains as an assembly and re-export platform ([Fajgelbaum et al., 2024](#)), making Vietnamese and Chinese exports to the US close substitutes at the product level and warranting an elasticity above the cross-origin average. For the ROW aggregate, $\sigma = 2$ reflects the attenuation of bilateral substitution patterns by aggregation across a large and heterogeneous set of trading partners ([Fajgelbaum et al., 2020](#); [Feenstra et al., 2018](#)).

D Extended Calibration Results

This appendix presents calibration parameters and model-vs-data comparisons for eight third-country configurations: the European Union and Vietnam ([under the same parameter values as the main text](#)), plus Japan, South Korea, Mexico, Canada, Taiwan, and India. For each configuration, σ is assigned based on the nature of the bilateral trading relationship with China in US import markets. EU and Vietnam are included for completeness alongside the additional configurations. All data are sourced as described in Appendix C; observed exchange rate changes are percent changes relative to the 2024 annual average.

The broad pattern of fit mirrors what is discussed for the three main configurations in Section 4. Under Regime 1 the model predicts dollar appreciation against both the RMB and every third-country currency, concentrated on the RMB; the data match the direction against the RMB and most third currencies, with the model overpredicting bilateral USD/RMB magnitudes at the one-month horizon. Under Regime 2 the model predicts modest dollar appreciation against the RMB (matching the observed -1.51 percent in sign) in all configurations except Canada, alongside USD depreciation against every third-country currency, consistent with the trade-war mechanism. The data confirm the third-currency depreciation for the euro and the yen; for most other trading partners the financial risk-off channel, which the model excludes, dominates at the April 2025 horizon, and the model’s predicted depreciation magnitudes for small open economies (Canada, Taiwan, India) remain far too large, reflecting the counterfactual home bias implied by world-export-share preference weights, as discussed in Section 4. South Korea and Taiwan follow the same pattern as the other non-safe-haven currencies; their divergence reflects their deep integration in US-oriented trade and supply chains rather than any emerging-market classification.

Table 4: Calibration parameters by third-country configuration. $\alpha_N = 0.60$ throughout. Tradable shares are constructed from 2024 WTO world goods export shares; labor endowments from IMF WEO 2024 PPP GDP ($L_A = 1$). σ assigned as described in main text and notes below.

Country C	σ	$\alpha_{T,A}$	$\alpha_{T,B}$	$\alpha_{T,C}$	α_N	L_C/L_A
European Union	6	0.097	0.165	0.138	0.600	0.860
Japan	4	0.129	0.220	0.051	0.600	0.226
South Korea	5	0.132	0.225	0.043	0.600	0.103
Mexico	5	0.132	0.224	0.044	0.600	0.113
Canada	2	0.134	0.228	0.038	0.600	0.086
Vietnam	8	0.138	0.235	0.026	0.600	0.055
Taiwan	3	0.135	0.230	0.035	0.600	0.058
India	3	0.137	0.233	0.031	0.600	0.497

Notes: σ values: EU ($\sigma = 6$) anchored on Amiti, Redding & Weinstein (2019) cross-origin central estimate; Japan ($\sigma = 4$) advanced autos and machinery with limited Chinese overlap; South Korea ($\sigma = 5$) and Mexico ($\sigma = 5$) meaningful product-level overlap in electronics and manufacturing; Canada ($\sigma = 2$) commodity-heavy exports (energy, mining) with low substitutability; Vietnam ($\sigma = 8$) deep supply-chain integration with China; Taiwan ($\sigma = 3$) semiconductor-dominant with limited aggregate substitutability; India ($\sigma = 3$) growing manufacturing base with limited current overlap.

Table 5: Model-implied and realized exchange rate changes by regime. Model columns report percent changes relative to the free-trade benchmark; data columns report percent changes relative to the 2024 annual average. Δe_{AB} : USD/RMB; Δe_{AC} : USD/ C -currency; Δe_{BC} : RMB/ C -currency. Positive values denote depreciation of the first currency. [Model values are computed under the corrected demand system of Section 3 \(realized-share tariff revenue and share-linear weights\).](#)

Country C	Model (%)			Data (%)		
	Δe_{AB}	Δe_{AC}	Δe_{BC}	Δe_{AB}	Δe_{AC}	Δe_{BC}
<i>Panel A: Regime 1 (Fentanyl-related tariffs, March 2025)</i>						
European Union	-5.39	-0.44	+5.23	-0.91	-0.23	+0.69
Japan	-7.95	-1.14	+7.40	-0.91	+1.42	+2.35
South Korea	-8.27	-0.92	+8.02	-0.91	-6.46	-5.60
Mexico	-8.22	-0.91	+7.96	-0.91	-10.00	-9.17
Canada	-9.60	-2.80	+7.52	-0.91	-4.61	-3.73
Vietnam	-8.27	-0.49	+8.48	-0.91	-1.93	-1.03
Taiwan	-9.04	-1.79	+7.98	-0.91	-2.73	-1.84
India	-7.91	-1.55	+6.91	-0.91	-3.43	-2.54
<i>Panel B: Regime 2 (Peak escalation, April 2025)</i>						
European Union	-2.50	+3.57	+6.23	-1.51	+3.36	+4.94
Japan	-1.57	+9.97	+11.72	-1.51	+4.55	+6.15
South Korea	-2.26	+7.59	+10.08	-1.51	-5.66	-4.21
Mexico	-2.27	+7.55	+10.04	-1.51	-9.33	-7.94
Canada	+1.12	+16.87	+15.58	-1.51	-2.29	-0.79
Vietnam	-3.39	+2.42	+6.02	-1.51	-3.06	-1.57
Taiwan	-0.27	+14.62	+14.92	-1.51	-2.00	-0.50
India	-0.58	+12.67	+13.33	-1.51	-2.31	-0.81